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MASTER'S DEGREE IN PHYSICS OF COMPLEX SYSTEMS

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and compares them with stochastic transport models.

This document reports the current state of the project. A first, reproducible acquisition has collected 212,602 flights from the FFVL Coupe Fédérale de Distance across two disciplines: 203,343 paraglider flights (seasons 1999–2000 to 2025–2026) and 9,259 hang-glider flights (2001–2002 to 2025–2026). Of these, 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified. Data collection is ongoing and may be extended to further public soaring databases – notably WeGlide for sailplanes, and XContest – using the same approach. The remainder of the document describes the acquisition method, the dataset and its statistics, and the planned analysis and modelling.

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Objectives	3
1.3	Scope of this document	4
2	The dataset	5
2.1	Data acquisition	5
2.1.1	Source: the FFVL Coupe Fédérale de Distance	5
2.1.2	What is collected	5
2.1.3	Pipeline and tooling	5
2.1.4	Reproducibility	6
2.1.5	Integrity	6
2.1.6	Further data sources	6
2.2	The IGC tracklog format	6
2.3	Flight metadata and the catalog	7
2.4	Glider categories	8
2.5	Coverage and statistics	8
2.6	Data quality and caveats	10
2.7	Trajectory pre-processing	10
2.7.1	Choice of altitude channel	11
2.7.2	Fix-level cleaning	13
2.7.3	Trimming of ground phases	18
2.7.4	Flight-level filtering	18
2.7.5	From geographic to local Cartesian coordinates	19
2.7.6	Uniform sampling within a flight	22
2.7.7	Smoothing and differentiation	24
2.7.8	Notation	27
2.8	Preliminary characterization	27
3	Next steps	29
3.1	Observables on the whole trajectory	29
3.2	Segmentation into flight phases	32
3.3	Phase-resolved observables	33
3.4	Reproducing the reference results	34
3.5	Transport analysis	34
3.6	Modelling and simulation	34
3.7	Tooling	35
A	CTRW with power-law waiting times: EA-MSD vs TA-MSD	38
B	Power spectral density and Welch’s method	41

Implementation and Computational Details	42
1 Overview	44
2 The dataset	46
2.1 Data acquisition	46
2.1.1 Pipeline and tooling	46
2.1.2 Reproducibility	46
2.1.3 Integrity	47
2.2 The IGC tracklog format	47
2.3 Flight metadata and the catalog	47
2.4 Glider categories	47
2.5 Coverage and statistics	47
2.6 Trajectory pre-processing	47
2.6.1 Choice of altitude channel	47
2.6.2 Fix-level cleaning	49
2.6.3 Trimming of ground phases	54
2.6.4 Flight-level filtering	54
2.6.5 From geographic to local Cartesian coordinates	54
2.6.6 Uniform sampling within a flight	55
2.6.7 Smoothing and differentiation	56
2.7 Preliminary characterization	56
3 Next steps	57
3.1 Observables on the whole trajectory	57
3.2 Segmentation into flight phases	57
3.3 Phase-resolved observables	57
3.4 Reproducing the reference results	57
3.5 Transport analysis	57
3.6 Modelling and simulation	57
3.7 Tooling	57

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from the moving atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight, energy-efficient path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is the setting in which *anomalous transport* arises. Let $\mathbf{r}(t)$ denote the horizontal position of a flight, measured from take-off. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle |\mathbf{r}(t)|^2 \rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. Continuous-time random walks and Lévy walks are the standard frameworks for such regimes [1,2], which have been reported in animal movement, in disordered media, and in turbulent flows. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [3]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [4], who analysed a large set of human soaring flights, found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$), and introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on that study: it enlarges the dataset and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues three objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;
2. to reconstruct clean trajectories from the raw GPS logs and define suitable transport observables;

3. to test for anomalous transport and to compare the empirical statistics with stochastic transport models.

A first dataset—the FFVL CFD archive of paraglider and hang-glider flights—has been assembled and verified and is described here; data collection is ongoing and may incorporate further sources, notably sailplanes (Sec. 2.1). The second and third objectives define the work ahead (Chapter 3).

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. Its quantitative content—the summary statistics and tables—is generated directly from the dataset, so it remains consistent with the data in hand. Chapter 2 follows the data end to end: the acquisition and the reproducibility of its pipeline (Sec. 2.1), the dataset and its statistics, and the pre-processing that turns the raw logs into analysis-ready trajectories. Chapter 3 sets out the planned transport analysis and modelling. Throughout, the discussion keeps to the scientific and statistical reasoning behind each choice. Implementation-level detail (code modules, configuration values, sample sizes, computation times, reproduction commands) is collected separately in Implementation and Computational Details, from p. 44, organised chapter by chapter to mirror the body and referenced at the relevant point as “implementation details: Sec. X”.

Chapter 2

The dataset

This chapter follows the data end to end: how the raw archive is acquired, verified and kept reproducible (Sec. 2.1); what the records contain and what the collection looks like (Secs. 2.2–2.6); and the pre-processing that turns a raw tracklog into the analysis-ready trajectory the transport analysis consumes (Sec. 2.7).

2.1 Data acquisition

2.1.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). The federation operates two parallel competitions on separate sites: one for paragliders (`parapente.ffvl.fr`) and one for hang gliders (`delta.ffvl.fr`). Pilots upload their flights; each entry carries metadata (pilot, date, take-off and landing sites, scored distance, duration, glider) and, in most cases, the raw GPS tracklog in IGC format. Both sites expose identical URL structure and machine-readable XML exports, so the same acquisition pipeline handles both. A “season” is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season; the paraglider archive starts in 1999–2000 and the hang-glider archive in 2001–2002.

2.1.2 What is collected

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, metadata plus the local path of its track) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository; only the small per-season index is versioned, and it is the source of the statistics of Sec. 2.5.

2.1.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package, exposed through two command-line tools, one per discipline, sharing the same subcommands for fetching, downloading, cataloguing, and verifying. The design favours robustness for a long-running, large job: downloads are resumable and safe to interrupt, a response is accepted only once validated as

real IGC content, and the client is courteous to the server it downloads from (implementation details: Sec. 2.1.1).

2.1.4 Reproducibility

The destination directory is not hard-coded: it is set per machine through an environment variable or a configuration file shipping a placeholder, so no machine-specific path is committed. Every command validates this setting at start-up and aborts with an explicit message if the target is missing, rather than failing obscurely. Archiving the season XMLs verbatim keeps the full provenance: the catalog can always be rebuilt from them (implementation details: Sec. 2.1.2).

2.1.5 Integrity

Because the data sit on an external drive, integrity is re-checked independently after download, rather than trusted at download time (implementation details: Sec. 2.1.3). Both FFVL collections have been verified in full: all 186,052 paraglider and 6,716 hang-glider downloaded tracklogs pass validation against the IGC standard (Sec. 2.2).

2.1.6 Further data sources

The FFVL CFD is the primary source of the present study, but the dataset is not assumed to be final. The paraglider data (203,343 flights, 186,225 tracklogs) are the main analysis target; the hang-glider data (9,259 flights, 6,746 tracklogs) are collected in parallel and enable the cross-type comparison of Sec. 2.4. Additional public soaring repositories are being pursued to enlarge and diversify the sample and, above all, to add the third glider category, *sailplanes* (Section 2.4). The most complete sailplane source is *WeGlide*: it exposes flight metadata and IGC tracks through an API, but the default access is rate-limited (of order sixty requests per day), so a bulk export of the archive is not feasible under it; a request for research access with a higher quota has been sent and is pending. Paraglider and hang-glider coverage may likewise be broadened through further portals such as XContest. These sources expose comparable flight metadata and IGC tracks and can be acquired with the same approach and merged into the same catalog.

2.2 The IGC tracklog format

A flight track is stored as an *IGC file*, the plain-text format defined by the FAI/IGC Technical Specification for IGC-approved GNSS flight recorders [5]; it is the standard output of the recorders used in free-flight competitions. Its layout is therefore fixed by that standard, not inferred from the files, and the acquisition validates every downloaded track against it (Sec. 2.1), so conformance is checked across the whole archive.

Each line is a *record* whose type is set by its first character – a code assigned by the standard, not chosen by the logger. The records relevant here are:

- A** the flight-recorder/manufacturer identifier (first line of the file);
- H** header records (flight date, pilot, glider, geodetic datum, ...);
- I** the declaration of the optional extra fields appended to every fix;
- B** the GPS fixes – the body of the file.

A B record stores its fields at fixed character positions. Decoded against the standard, the example

reads as: UTC time 09:43:29; latitude 49°33.044' N and longitude 000°05.193' E on the WGS84 datum; a fix-validity flag; and *two* altitudes in metres – the barometric (pressure) altitude (95 m) followed by the GNSS altitude (149 m).

The validity flag is fixed by the standard to two values and certifies the *GNSS altitude* rather than the horizontal position: **A** marks a valid three-dimensional fix (usable GNSS altitude), whereas **V** marks a two-dimensional fix or a GNSS drop-out, in which case the standard requires the GNSS-altitude field to be written as zero. A **V** fix can thus still carry a usable latitude/longitude and a valid barometric altitude.

Decoding a **B** record from fixed character positions can succeed numerically while the result is not a valid encoding at all – a corrupted byte can still parse as a digit. The parser therefore checks, at decode time and independently of any flight-dynamics judgement, that the record is actually a legal encoding – a valid time, valid latitude/longitude fields, a valid hemisphere letter – dropping it otherwise, on the same footing as a record with the wrong length or a non-numeric field (implementation details: Sec. 2.2). This is a check of *format* validity, distinct from the *plausibility* checks of fix-level cleaning (Sec. 2.7.2, e.g. the inter-fix speed bound), which target fixes that decode validly but would be physically implausible given the rest of the trajectory.

The two altitude channels differ in reference and in noise: the barometric value is referenced to the ICAO standard atmosphere and is smooth, the GNSS value is referenced to the WGS84 ellipsoid and is vertically noisier. On older loggers one channel may be absent (recorded as zero). The analysis adopts the *barometric* channel for the vertical dynamics, for the signal-quality reasons set out and quantified in Sec. 2.7.1; this choice matters because the vertical velocity derived from it drives the phase segmentation.

The sequence of **B** records is thus a time-stamped three-dimensional trajectory. The sampling period is typically a few seconds, is not guaranteed to be uniform, and varies across recording devices – a point the analysis must handle explicitly.

2.3 Flight metadata and the catalog

Every flight in the season XML carries a set of metadata as record attributes, which the acquisition preserves. Beyond the identifiers and links, the fields used downstream are: the **date**; the **pilot** and **club**; the **take-off** and **landing** sites and the French **department**; the glider, given as both a model name (**aile**) and a class (**aile_class**, Section 2.4); and the competition quantities – the **flight type** (the *scoring* category: free distance, out-and-return, flat or FAI triangle, hike-and-fly, ...), the scored **distance** and **points**, and, where present, the **duration** and average **speed**. The flight type is a scoring label and must not be read as a glider or manoeuvre type.

From these, the pipeline builds a single derived table, the **catalog** (**catalog.csv**), with *one row per flight* across all seasons: the metadata above plus a **local_path** linking each flight to its track on disk and a **downloaded** flag. It is the working index of the project – a flight can be traced to its file, and back (through the identifier embedded in the file name) to its metadata and public page, without any external lookup. A smaller **season index** (**seasons_index.csv**), versioned in the repository, summarises each season by the number of flights declared, carrying a track, and downloaded; it is the source of the statistics below. Both tables are regenerated from the archived XMLs on demand (**build-catalog**).

2.4 Glider categories

Soaring kinematics depend strongly on the aircraft, so the glider must be identifiable for any stratified analysis. Three broad categories exist, in increasing order of performance: **paragliders** (flexible canopy; slowest, glide ratio ~ 8 –11, tight turns), **hang gliders** (rigid-framed delta wing; faster, glide ratio ~ 10 –15), and **sailplanes** (rigid aircraft; fastest, glide ratio $\gtrsim 25$). They differ several-fold in cruise speed and turn radius – which is why the reference study [4] reports its transport statistics separately for each.

In practice the category is set by the *data source*, since the public databases are organised by discipline. The FFVL runs the Coupe Fédérale de Distance as two parallel competitions (one for *paragliders*, one for *hang gliders*), so the present dataset spans *both* glider types: the glider-type stratum has two levels, and the cross-type comparison of the reference study [4] is therefore already possible between paragliders and hang gliders. The third category, *sailplanes*, is collected by different portals and is not yet in hand (Sec. 2.1); until it is added the cross-type analysis is limited to the two FFVL disciplines.

Within each discipline the flights can be further stratified by *performance class*, recorded in `aile_class`; the two disciplines use different class systems. FFVL labels *paragliders* on the EN/AFNOR certification ladder – EN A (“A ou 1”, most stable, recreational), EN B (“B ou 1-2”), EN C (“C ou 2”), EN D (“D ou 2-3”), the “CIVL Competition Class” (CCC) racing wings, plus tandem (“biplace”) and non-certified (“non homologuée”) entries. *Hang gliders* follow the FAI/CIVL class system, which in this data reduces to flexible wings (“Delta Class 1” and the intermediate “Delta Class Sport”) and rigid wings (“Rigide Class 2” and “Rigide Class 5”). In both cases the ladder tracks performance (higher classes fly faster and flatter), so it is the natural within-source stratification variable, playing the role that glider type plays across sources.

Two caveats bear on any use of `aile_class`. First, the field must be normalised before it can serve as a stratum, since a small fraction of rows carry a blank or placeholder class. Second, the class certifies the wing, not the pilot. Both points are handled at the pre-processing stage, later in this chapter (implementation details: Sec. 2.4).

2.5 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Sec. 2.1). It comprises 212,602 flights in total, of which 192,971 carry a GPS tracklog and 192,768 have been downloaded and verified, split across two disciplines.

The *paraglider* collection spans 27 seasons, from 1999–2000 to 2025–2026: 203,343 flights, of which 186,225 (91.6 %) carry a tracklog and 186,052 (99.9 %) are downloaded and verified (Table 2.1). The *hang-glider* collection spans 25 seasons, from 2001–2002 to 2025–2026: 9,259 flights, of which 6,746 (72.9 %) carry a tracklog and 6,716 (99.6 %) are downloaded and verified (Table 2.2). The hang-glider archive is roughly twenty times smaller, reflecting the smaller active population of the discipline.

In both disciplines the volume of flights grows by orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute thousands of tracks for paragliders and several hundred for hang gliders.

Table 2.1: FFVL CFD *paraglider* flights per season (`parapente.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 2.5)

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

Table 2.2: FFVL CFD *hang-glider* flights per season (`delta.ffvl.fr`): total declared, with a GPS track, and downloaded. (implementation details: Sec. 2.5)

Season	Flights	With GPS	Downloaded	Coverage (%)
2001–2002	248	0	0	0.0
2002–2003	281	6	6	100.0
2003–2004	256	28	25	89.3
2004–2005	244	35	25	71.4
2005–2006	365	80	76	95.0
2006–2007	337	100	96	96.0
2007–2008	388	136	132	97.1
2008–2009	344	155	155	100.0
2009–2010	324	136	136	100.0
2010–2011	425	147	147	100.0
2011–2012	117	103	103	100.0

Season	Flights	With GPS	Downloaded	Coverage (%)
2012–2013	428	406	406	100.0
2013–2014	423	409	409	100.0
2014–2015	628	595	591	99.3
2015–2016	417	413	413	100.0
2016–2017	468	462	462	100.0
2017–2018	342	331	331	100.0
2018–2019	604	594	594	100.0
2019–2020	378	373	373	100.0
2020–2021	328	328	328	100.0
2021–2022	570	568	568	100.0
2022–2023	376	374	374	100.0
2023–2024	437	437	436	99.8
2024–2025	353	352	352	100.0
2025–2026	178	178	178	100.0
Total	9,259	6,746	6,716	99.6

2.6 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of paraglider entries and about 27 % of hang-glider entries have no IGC file (older or manually declared flights); they carry metadata only.
- **A small residue of unrecoverable files.** A few flights advertise a track that the server does not actually serve as a valid IGC (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, altitude source, and accuracy depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Altitude channel.** The barometric channel is adopted for the vertical dynamics (Sec. 2.7.1), but a sizeable minority of loggers record no pressure altitude – over a quarter of paraglider and roughly a sixth of hang-glider flights – which fall back to the GNSS altitude, flagged as such.
- **Glider metadata.** Flights are paragliders or hang gliders, tagged by source (Section 2.4); the class field `aile_class` uses a different system in each discipline and must be normalised before it is used to stratify.
- **Incomplete dates.** Some historical entries have placeholder dates (e.g. 0000-00-00); these are kept but flagged.

These caveats motivate the pre-processing described in the remainder of this chapter.

2.7 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – UTC time, geodetic latitude and longitude, and altitude – by the `soaring.analysis.igc` parser, which reads the **B** records at their fixed character positions and returns both altitude channels. The pipeline then proceeds in a fixed order:

- (i) choose the altitude channel (Sec. 2.7.1);
- (ii) clean individual fixes (Sec. 2.7.2);
- (iii) trim the ground phases (Sec. 2.7.3);
- (iv) filter out unfit whole flights (Sec. 2.7.4);
- (v) convert to a local Cartesian frame (Sec. 2.7.5);
- (vi) enforce a uniform sampling interval within each flight (Sec. 2.7.6);
- (vii) smooth and differentiate the trajectory (Sec. 2.7.7).

The output is a tidy, analysis-ready representation in a growing `soaring.analysis` sub-package.

The ordering is chosen so that each step runs in the frame where it is exact and cheap. Steps (i)–(iv) act on the *raw geographic* coordinates, since the only geometric quantities they need (inter-fix speed and spatial extent) follow from the great-circle (haversine) distance between fixes, with no coordinate conversion. The conversion (v) to the metric East–North–Up frame (Sec. 2.7.5) is then done on the cleaned, trimmed track, with the origin at the take-off fix ($\mathbf{r}(0) = \mathbf{0}$), so the tangent-plane frame is built from good data and referred to the right origin. The final steps (vi)–(vii), which produce filtered position, velocity and acceleration, are naturally metric and so run *after* the conversion.

2.7.1 Choice of altitude channel

The vertical dynamics – vertical velocity, climb/sink, and the phase segmentation that rests on them – are taken from the *barometric* altitude rather than the GNSS altitude. The reason is one of signal quality in the frequency band that matters. The barometric (pressure) altitude has sub-metre relative precision and very little high-frequency noise; its errors are slow: a constant offset from the ICAO reference, tens of metres up to order 100 m (read directly in Fig. 2.1a), and a drift of tens of metres over a multi-hour flight, following the synoptic pressure and the atmosphere’s departure from the ICAO profile (Fig. 2.1b). Slow errors are annihilated by the differencing that every dynamical quantity involves (a vertical velocity, a displacement increment, an autocorrelation). The GNSS altitude, by contrast, is geometrically referenced but vertically noisier: the vertical dilution of precision makes it 1.5–2 times worse than the horizontal, and multipath adds high-frequency jitter of a few metres, exactly the band that differencing amplifies. This is why free-flight variometers derive the climb rate from the pressure sensor rather than from GNSS.

Empirical confirmation. This is borne out on our data, with one qualification. Figure 2.1 compares the two channels over a subsample sized for a robust spectral estimate (Appendix B) (implementation details: Sec. 2.6.1). At low frequency the two agree, tracking the real climbs and glides. The high-frequency behaviour is where they differ, though not uniformly. For the *typical* flight the two are comparable: both approach the common floor set by the metre resolution to which the IGC format rounds *both* channels, so the median spectra (panel c, solid lines) nearly coincide.

What separates the channels is the *spread* across flights, the shaded bands. The barometric band stays tight (that channel is uniformly clean), whereas the GNSS band fans out by one to two orders of magnitude towards the Nyquist frequency: a substantial minority of flights carry large GNSS high-frequency noise, from multipath and the vertical dilution of precision, while most do not. Since the vertical velocity is a finite difference, which amplifies exactly that band, and since one cannot know in advance which flights have poor GNSS reception, the uniformly clean barometric channel is the safe choice. This *departs from the reference*

study [4], whose pipeline derives the vertical coordinate and velocity from the GNSS altitude; the divergence is deliberate, and one of the points re-examined here.

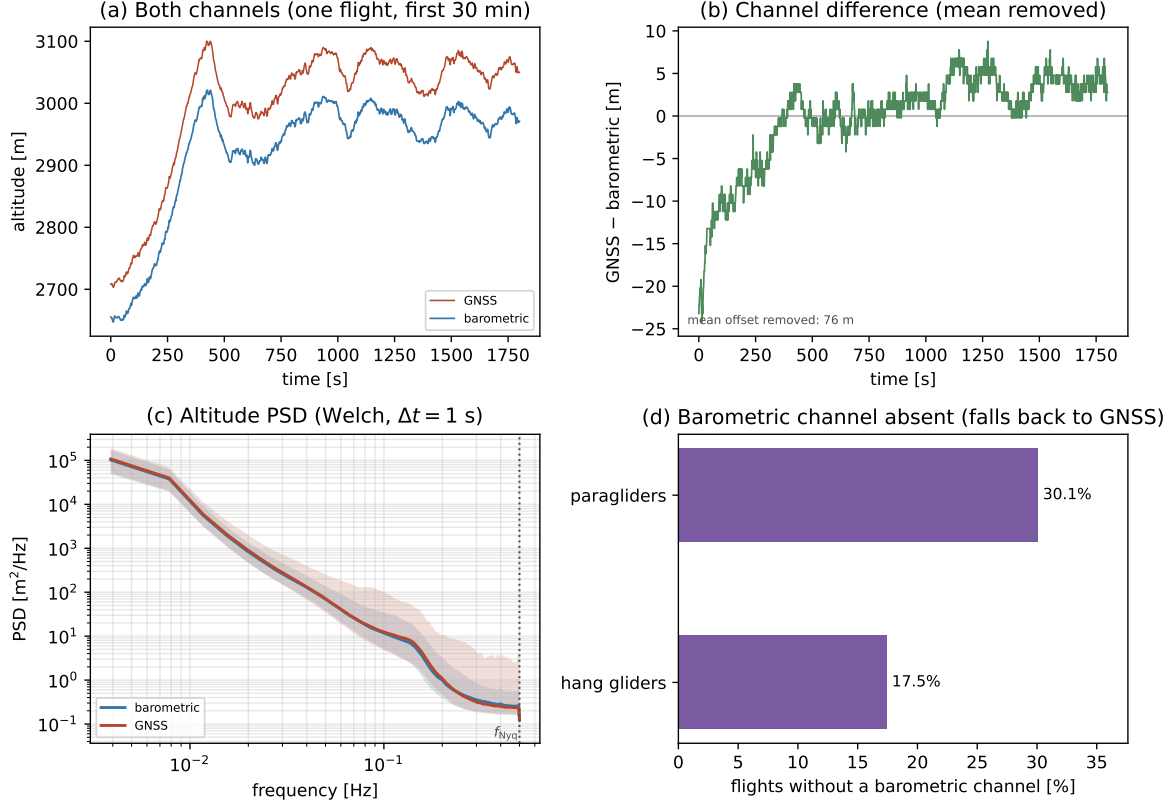


Figure 2.1: Barometric vs GNSS altitude noise. Panels (a)/(b): one representative flight over a 30 min window. (a) the two raw channels on a single *shared* axis, so their offset – the gap between the pressure and ellipsoidal references, tens of metres – is read directly, while both track the same vertical motion. (b) their difference GNSS–barometric (mean removed, *not* detrended): the offset drifts by tens of metres over the flight, a real feature – plausibly the barometric channel’s slow departure from the ICAO standard atmosphere as altitude changes – not noise, so only its constant part (annotated) is subtracted. Panels (c)/(d): over the ensemble. (c) the per-frequency *median* Welch spectrum of each channel (Appendix B), pooled over the two disciplines (whose spectra coincide), with a shaded 10 to 90 percentile band across flights. The median lines nearly coincide: for the *typical* flight the two channels are comparable at high frequency, both near the floor set by the shared metre-resolution logging. The relevant difference is in the *bands*: the barometric band stays tight (uniformly clean), whereas the GNSS band fans out by one to two orders of magnitude towards Nyquist, driven by a minority of flights with heavy multipath noise. The median is used rather than the mean because these per-flight spectra are heavy-tailed and a mean is dominated by that noisy minority (Sec. 2.6.1). The dotted line is the Nyquist frequency $f_{\text{Nyq}} = 1/(2\Delta t)$. (d) Fraction of flights whose barometric channel is absent, per discipline. (implementation details: Sec. 2.6.1)

Several practical points follow.

One channel per flight. A trajectory uses a single altitude channel end-to-end; the barometric and GNSS values are never spliced within a flight, since a mid-flight switch between two channels offset by tens of metres would inject a spurious jump.

Source recorded. Which channel a flight uses is stored as a per-flight field – an `alt_source`

column (**baro** or **gnss**) in the processed dataset, alongside duration, fix count and the like; the raw IGC files are never modified.

Availability. A flight is used with its barometric channel when it has one, and the few individual fixes with a missing barometric value are dropped rather than back-filled from GNSS. A logger with *no* pressure sensor writes the whole barometric channel as zero (the presence is essentially all-or-nothing, not a scattering of gaps); such a flight falls back to the GNSS altitude, unfiltered – a non-negligible minority of flights (Figure 2.1d). Presence alone is not enough: a stuck sensor writes a constant non-zero value and would feed the segmentation a vertical velocity of identically zero, so a barometric channel whose total range over the flight stays below 30 m is treated as absent, and the flight falls back to GNSS.

Measuring the fallback rate. Because it is a headline number, its precision is exact rather than estimated: both archives are full-censused – 17.5 % of the 6,716 hang-glider and 30.1 % of the 186,025 paraglider flights scanned carry no barometric channel (implementation details: Sec. 2.6.1). (The counts quote the census scan directly, through generated macros; the archive keeps growing by a trickle, so they can trail the live download totals of Sec. 2.5 slightly until the scan is refreshed.)

Mixed altitude sources. The mix is acceptable, for two reasons. The *horizontal* transport analysis, the primary result (MSD and its scaling), does not use the altitude at all, so every flight is usable there regardless of channel. Altitude enters only the vertical velocity, hence the segmentation and the phase-resolved observables; there the two source groups are kept distinct through **alt_source** and their statistics checked for compatibility (Sec. 2.8). If the noisier GNSS-only flights were to bias a vertical statistic, that analysis is restricted to the barometric flights – still the large majority – while the horizontal analyses keep the full sample. Forcing GNSS on *all* flights, to avoid the mix, would be the wrong trade: it would needlessly degrade the vertical signal of the barometric majority to match the worst loggers.

2.7.2 Fix-level cleaning

Geometry on the raw coordinates. Cleaning, trimming and flight-level filtering all run *before* the Cartesian conversion (Sec. 2.7.5), directly on the geodetic angles (φ_i, λ_i) . Distances between fixes therefore cannot be Euclidean; they are great-circle (haversine) distances on a sphere of mean radius $R_\oplus = 6,371$ km. For two consecutive fixes,

$$\Delta s_i = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\varphi_{i+1} - \varphi_i}{2} + \cos \varphi_i \cos \varphi_{i+1} \sin^2 \frac{\lambda_{i+1} - \lambda_i}{2}}, \quad (2.1)$$

from which the horizontal speed between the fixes is $v_{xy,i} = \Delta s_i / (t_{i+1} - t_i)$ and, from the barometric altitude, the vertical speed is $v_{z,i} = (h_{i+1} - h_i) / (t_{i+1} - t_i)$. Two whole-flight quantities used below follow from the same distance: the flown *path length* $L = \sum_i \Delta s_i$ and the *extent* $D = \max_i \Delta s(\text{fix}_0, \text{fix}_i)$, the farthest a fix gets from take-off. The spherical approximation is ample here (the ellipsoidal correction to an inter-fix distance is $\lesssim 0.3$ %, far below the GPS noise), and it avoids converting coordinates before the trajectory has been cleaned.

Design principle. Corrupt or physically impossible fixes must be removed without discarding the flight around them. At the same time, every removed or altered fix modifies the trajectory whose statistics are about to be measured, and an over-aggressive cleaning biases the result

as much as dirty data would. The cleaning therefore assumes as little as possible about the motion: only smoothness at the sampling scale (an aircraft cannot jump between fixes a few seconds apart), never a model of the transport itself. A model-based smoother would reject outliers more effectively, but fitting a constant-velocity or diffusive model to the track would imprint the behaviour under study onto the cleaned data.

The two possible errors have different costs, and the cleaning treats them asymmetrically. The slow, localized phases of a flight (search and climb) correspond to the *waiting times* of the transport model (Ch. 3), and their durations populate the tail of $\psi(\tau)$, where the anomalous exponent is determined. A spurious fix left in a slow phase has a bounded effect: to survive the tests below it must lie within tens of metres of the true track (at most the speed bound times the native step, ≈ 45 m at 1 Hz, and typically within the outlier floor $\varepsilon_{\min} \approx 15$ m), a displacement the smoothing absorbs. Removing a genuine slow-phase fix, by contrast, erases a real trapping event, and nothing downstream can restore it.

The cleaning therefore defaults to keeping the data: a fix is removed only on positive evidence of a defect, and no detector deletes a fix that still carries a usable horizontal position (at most, a corrupt channel is marked missing and restored downstream).

Three classes of defect. Past the format-validity check made at decode time (Sec. 2.2), which rejects any record that is not a legal encoding, a surviving fix can still be wrong in one of three ways, summarized in Table 2.3. A fix *off the trend* – a GPS position spike, or a jump out and back within one step – is found by a robust local-outlier test – the *Hampel identifier* [6]: its residual r_k from a median estimate built on the fixes around it is scaled by the robust local spread of those residuals, and the fix is flagged when

$$r_k > \max(k \sigma_k, \varepsilon_{\min}), \quad \sigma_k = 1.4826 \text{MAD}_k, \quad (2.2)$$

i.e. when it departs by more than k robust standard deviations *and* by more than an absolute floor $\varepsilon_{\min}$ of a few times the GPS noise (working defaults: $k \approx 4\text{--}5$, a window $\pm w$ with $w \approx 15$ s to 30 s, $\varepsilon_{\min} \approx 15$ m; rationale in Sec. 2.6.2). A flag alone does not delete: the fix is removed only when this local flag is *corroborated* by an absolute v_{xy} bound that rules the step physically impossible, so a genuine but under-sampled tight turn – flagged, yet flying at an ordinary speed – is kept (Sec. 2.6.2). A fix *on the trend* agrees with its neighbours yet is still wrong as a record – a duplicated or backward timestamp, or a run of frozen positions from a lost GNSS lock; its residual is near zero, so it escapes the outlier test and needs a structural rule instead. A value *out of range*, an impossible altitude or climb rate, is caught by an absolute bound that needs no context. (Status: the decode-time format check is implemented and unit-tested, and the absolute bounds are thresholded on the data; the detectors themselves are designed but the routine that walks a trajectory applying them is still to be built – Sec. 2.6.2 tracks this precisely.)

Removal policy. A speed bound alone cannot attribute a defect: the speed belongs to the segment *between* two fixes and does not identify which endpoint is wrong. The local-outlier test resolves the attribution, because the residual is a property of a single fix; an isolated spike is removed without touching its neighbours, and a short burst is removed as one block, the smallest whose removal restores the trend. The action then depends on the channel affected. A corrupt *position or time* deletes the fix entirely; the value is not replaced with the local median, because all reconstruction is deferred to the single resampling step (Sec. 2.7.6), which bridges the resulting short gap. A defect that no bounded removal repairs – a position discontinuity, such as a re-acquisition offset, where the track jumps and stays – is not a deletion problem: both sides are genuine and only the transition between them is unknown, so the trajectory is split at the step, like a long gap (Sec. 2.7.6). A corrupt *altitude* keeps position and time and

marks only the altitude as missing; the horizontal analysis, which is the primary result, does not use the altitude, so dropping a barometric spike costs it nothing (Sec. 2.7.1).

Frozen-position runs (GNSS lock loss). When the receiver loses its GNSS lock, many loggers repeat the last position for tens of seconds while the clock keeps running. Such a run passes every test above: horizontal speed zero, altitude in range, valid timestamps. If kept, it enters the analysis as a long *waiting time* and contaminates the tail of $\psi(\tau)$; if the detector is too aggressive, however, it splits tight thermalling climbs, which are also slow and localized, and destroys real trapping events. The second error is the more damaging of the two, so a run is cut only when three independent signatures agree, combined as in Eq. (2.3) below:

- (i) *Spatial extent, not step.* A lost lock collapses the whole run into the logger’s quantization: *every* fix of the run lies within ε (a few times the GPS noise) of every other, so the run’s bounding diameter stays below ε . A circling climb instead traces a disc of diameter $2R \approx 40$ m to 100 m, far larger than ε , and clears the test on extent alone. The criterion is the run’s *extent*, never the fix-to-fix step: at 1 Hz a climbing wing advances ~ 10 m per fix, comparable to ε , so a step-based test would flag every climb.
- (ii) *An independent witness – the barometric altitude.* A GNSS lock-loss freezes only the GNSS position; the pressure sensor is a *separate* instrument. A wing that is really thermalling keeps gaining height on the barometric channel – a climb rate of order 1 m s^{-1} over the run is tens of metres – whereas a true freeze shows a flat barometric trace. A horizontally motionless run whose barometric altitude is still changing at a plausible rate is a (tight) climb, and is *kept* – unless its coordinates repeat byte-identically, the one signature allowed to overrule this witness (iii).
- (iii) *Recorder declarations.* Two signatures declare a defect directly, and they carry different weight. A *byte-identical* repeat of the coordinates is conclusive on its own: no moving receiver rewrites the very same bits, so the positions are invented whatever the barometer says. The V flag and the zeroed GNSS altitude are weaker: they certify a degraded *GNSS altitude*, not a frozen horizontal fix – the standard lets a V fix carry a usable position (Sec. 2.2) – so they decide only where no barometric witness exists, i.e. on a GNSS-fallback flight.

A run is classified as a loss of signal only when it is spatially collapsed, *witnessed*, and sufficiently long,

$$\underbrace{\text{diam}(\text{run}) < \varepsilon}_{\text{(i) collapsed}} \wedge \underbrace{W}_{\text{witness (ii)/(iii)}} \wedge \underbrace{\Delta t_{\text{run}} \geq \tau_{\text{freeze}}}_{\text{duration}}, \quad (2.3)$$

$$W = \begin{cases} |\Delta z_{\text{baro}}| < \delta_z \vee \text{byte-identical} & \text{barometric flight,} \\ \text{V/zero-alt} \vee \text{byte-identical} & \text{GNSS-fallback flight.} \end{cases}$$

The run is then marked as a gap and the trajectory is *split* there rather than interpolated across (Sec. 2.7.6), so that no motion is invented over an interval whose true path is unknown. The witness W is hierarchical. On a barometric flight no declaration can overrule a climbing barometer, because a V flag certifies the GNSS altitude and not the horizontal fix (Sec. 2.2); the one exception is the byte-identical repeat, which is conclusive on its own. Coordinates repeated bit for bit are invented positions even when the height gain is real, so such a run is cut even while the barometric altitude climbs, and only the wind drift the receiver never recorded is lost.

Quantitatively, a real thermalling climb fails the cut on two independent instruments at once: its circle spans $2R \approx 40$ m to 100 m against $\varepsilon \approx 10$ m to 15 m, so test (i) misses by a

factor of 3–10; and a climb of order 1 m s^{-1} sustained over $\tau_{\text{freeze}} \gtrsim 60 \text{ s}$ to 90 s moves the barometer by $\gtrsim 60 \text{ m}$ against a flatness tolerance δ_z of a few metres. Cutting a genuine climb on a barometric flight therefore requires two simultaneous order-of-magnitude failures, or coordinates repeated bit for bit, which no moving receiver writes.

The rule assumes a functioning barometric channel, and degrades where that channel is absent: on the GNSS-fallback minority (Sec. 2.7.1) the declared signatures decide, and both residual corner cases are confined there – a sub- ε circling climb carrying a V flag (cut wrongly) and an undeclared, jittering freeze (kept wrongly), each a near-empty set in which the default to keep favours preserving real dynamics. One benign miss remains on barometric flights: a freeze whose repeats are jittered rather than byte-identical under a climbing barometer is kept – at bounded cost, since within a sub- ε run the true position is nearly stationary, and only the unrecorded wind drift is lost.

Defect	Detected by	Action
Off the trend — <i>the horizontal position itself is corrupt</i>		
Position spike, out-and-back jump	robust local-outlier test (Hampel identifier [6]), <i>corroborated by</i> an absolute v_{xy} bound ($45/55 \text{ m s}^{-1}$, para/hang)	delete the fix; the short gap is bridged at resampling
Position discontinuity (re-acquisition offset)	flagged step that no bounded removal rejoins	split at the step, like a long gap
On the trend — <i>record structurally wrong, position agrees with its neighbours</i>		
Duplicate timestamp	two or more fixes in one UTC second	merge to one (centroid)
Backward timestamp	time decreases (not a midnight roll-over)	delete the fix
Frozen-lock run	diam $< \varepsilon$ for $\geq \tau_{\text{freeze}}$, witnessed (Eq. 2.3): baro. flat or byte-identical; V/zero-alt on GNSS-fallback	mark as gap, <i>split</i> there
Out of range — <i>position intact, only a sensor value is impossible</i>		
Missing / out-of-band altitude, or $ v_z $ spike	channel absent, outside $[-100, 6000] \text{ m}$, or baro. $ v_z > 13 \text{ m s}^{-1}$	drop the altitude only; the fix stays
Invariant: a usable horizontal position is never discarded. A fix is deleted only when its <i>own</i> position is corrupt or invented (frozen lock); when only a sensor value is wrong, the fix is kept and that channel is marked missing, to be restored at resampling (Sec. 2.7.6).		

Table 2.3: Fix-level cleaning, grouped by how each defect presents; the three groups map onto detectors of increasing context: an absolute bound, a robust local test, a structural rule. Numeric thresholds and the robust-test parameters (k , $\varepsilon_{\min}$, ε , τ_{freeze}) are audited in Fig. 2.2 and set in Sec. 2.6.2; the v_{xy} bound is per-discipline because the speed envelopes differ (a **sailplanes** entry follows once that source is in). The altitude lower bound is -100 m , not 0 , because the un-QNH-corrected barometric offset (Fig. 2.1) lets a sea-level take-off read slightly negative. (implementation details: Sec. 2.6.2)

The IGC validity flag. The IGC validity flag (A/V) certifies the GNSS altitude only, so its effect is already covered by the missing-altitude rule on the adopted channel. A V fix keeps a valid position and, where present, a valid barometric altitude (Sec. 2.2): on a barometric flight it is kept, on a GNSS-fallback flight it has no usable altitude and drops out as missing.

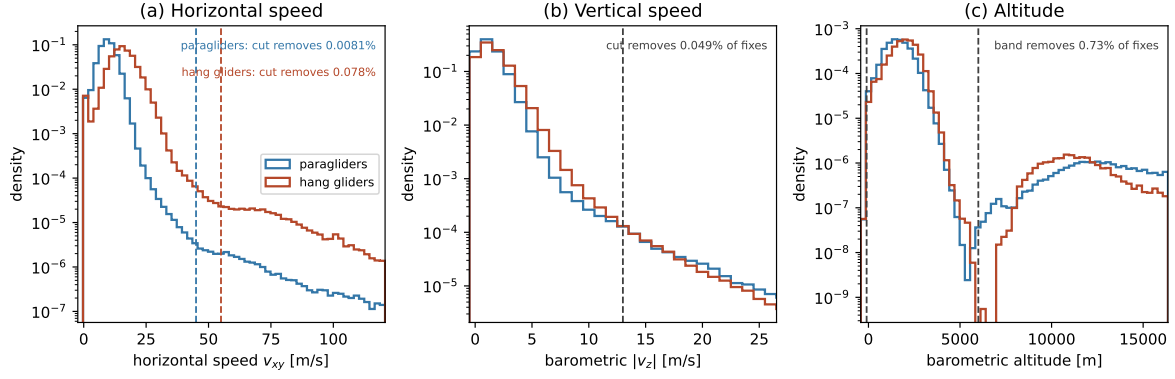


Figure 2.2: Fix-level cleaning: auditing the absolute bounds. Per-*fix* distributions of the quantities the out-of-range bounds test – (a) horizontal speed v_{xy} , (b) barometric $|v_z|$, (c) barometric altitude – pooled over a large paraglider subsample and the complete hang-glider population, with the adopted bounds dashed (Table 2.3); the y -axis is logarithmic so the sparse tail the bounds act on is visible. These are the *context-free* cuts, the floor on the plainly impossible; the robust local-outlier and frozen-lock tests act on structure a marginal distribution cannot show and are validated separately (Sec. 2.6.2). Panel (a) is why the speed bound is per-discipline: hang gliders show a distinct, non-decaying plateau from about 55 m s^{-1} to 90 m s^{-1} on top of the steeply decaying real distribution – a sustained *ground* speed of 200 km h^{-1} to 320 km h^{-1} has no airborne explanation for a hang glider and reads as a logger artefact; the bound is placed where the decay *stops*, not where it first slows, since a shallow but real decreasing stretch can still be genuine rare dynamics. Paragliders show the same pattern lower down. Panel (b) uses one bin per integer m s^{-1} : barometric altitude is logged to the metre, so a finite difference returns integer rates and a finer grid would alias them into a spurious comb; its 13 m s^{-1} bound sits above the strongest thermals and ordinary sink but below what a sustained spiral dive can reach, so the validation checks that the fixes it cuts are isolated, not run-shaped (Sec. 2.6.2). Panel (c) uses a narrower range, its cut sitting in an otherwise tight, unimodal distribution. (implementation details: Sec. 2.6.2)

Neither case needs handling of its own.

Completeness of the resampled trajectory. The segmentation (Ch. 3) needs a defined horizontal position *and* altitude at every time, since the vertical velocity is its discriminant. Cleaning does not guarantee this fix by fix – it may delete a node (a corrupt position) or mark a channel missing (a barometric spike, a V fix on a GNSS-fallback flight) – because the guarantee is established one step later, at resampling (Sec. 2.7.6): within each retained segment every grid time is given a complete (x, y, z) , the short holes filled by interpolation per channel and the long ones removed by splitting. The invariant is thus “complete (x, y, z) *within* a segment”, not across a split. This is why the validity flag, and the missing-altitude rule it feeds, need no repair of their own here: a dropped altitude is not a lost altitude but a *deferred* one, reconstructed downstream where a single, audited interpolation does all the filling.

Flight-level integrity gate. Cleaning acts fix by fix, but it also keeps count of how much it had to do. The counters are taken over the airborne window: defects in a ground phase that trimming removes anyway must not condemn the flight. If cleaning has removed or reconstructed more than a small fraction of a flight’s airborne fixes, the flight is no longer being cleaned but rebuilt, and it is dropped whole – the cleaning counterpart of the sampling-

regularity cut (Sec. 2.7.6).

Validating the cleaning. That each bound removes only a negligible fraction of fixes shows the cleaning is not too aggressive; it says nothing about what the cleaning *misses*. Two checks address that, and both belong to the planned implementation (Ch. 3). *Injected defects*: spikes, frozen runs and time glitches of known size are planted in clean tracks, and the fraction recovered as a function of size measures the false-negative rate directly. *Invariance of the result*: the transport estimators – the displacement-scaling exponent and the waiting-time tail – are recomputed as the cleaning is tightened, and the operating point is taken on the plateau where they stop moving. A cut that removes error and not signal leaves the physics unchanged; one that has begun to bite into real dynamics does not (implementation details: Sec. 2.6.2).

2.7.3 Trimming of ground phases

The logger starts recording before take-off and stops after landing; these ground phases must be stripped before any kinematic analysis. Detection uses the *horizontal* speed v_{xy} alone (from (2.1)) – the vertical speed plays no role here. With $v_0 = 5 \text{ m s}^{-1}$ and $T_0 = 30 \text{ s}$, the retained airborne segment is $[t_{\text{on}}, t_{\text{off}}]$ with

$$t_{\text{on}} = \min\{t : v_{xy} > v_0 \text{ on } [t, t + T_0]\}, \quad t_{\text{off}} = \max\{t : v_{xy} > v_0 \text{ on } [t - T_0, t]\}, \quad (2.4)$$

i.e. the same sustained-speed rule read forward and time-reversed – *not* a cut at the first sustained drop, which a slow soaring phase (ground speed near zero when hovering into wind) would otherwise trigger mid-flight. The persistence T_0 prevents a gust or a GPS glitch on the ground from being mistaken for take-off. A flight with no sustained stretch at all has no airborne segment and is discarded, with the reason recorded. The rule trims only the *outer* ground phases: a mid-flight landing and relaunch recorded in one log (a top-landing, say) would survive as an interior near-zero-speed stint and read downstream as an implausibly long wait – and a false wait of tens of minutes lands in the far tail of $\psi(\tau)$, where it does the most damage. An interior-ground detector is therefore specified, built like the frozen-lock rule: it cuts only on corroborated evidence, because its dangerous error is the opposite one – a wing soaring into wind can hold v_{xy} near zero for minutes while flying. A stint is cut only when both conditions hold: $v_{xy} < v_0$ sustained for at least T_{ground} (order ten minutes, far beyond any search), *and* the barometric altitude flat over the whole stint, within a few metres – a wing at zero ground speed in real air still moves vertically, the same independent-sensor argument as Eq. (2.3). On a GNSS-fallback flight the vertical witness uses the noisier GNSS channel with a wider tolerance, and in doubt the stint is kept. A cut stint is treated exactly like a long gap: excised and split (Sec. 2.7.6), inheriting the segment rules – boundary censoring, no second pass of the flight-level cuts. Stints shorter than T_{ground} are not cut but *flagged*: a wait with sustained $v_{xy} < v_0$ and a flat barometric trace is marked suspect whatever its length, and the $\psi(\tau)$ fits are re-run with and without the suspect waits as a sensitivity check – the same flag-without-deleting pattern as the outlier identifier. The parameters (T_{ground} and the flatness tolerance) join **trimming** in the configuration when the guard is built (implementation details: Sec. 2.6.3). One caveat is structural: the criterion is a *ground* speed, so a slow phase adjacent to take-off or landing (a launch straight into ridge lift, say) can still be clipped; the trimmed fraction is recorded per flight so this failure mode is auditable rather than hidden. Both the threshold and the persistence affect what fraction of each trajectory is retained, so they are made explicit (implementation details: Sec. 2.6.3).

2.7.4 Flight-level filtering

After cleaning and trimming, some flights are still unfit for the ensemble analysis and are dropped whole. The step is delicate: too aggressive a cut biases the transport statistics – a

high minimum duration, for instance, preferentially keeps successful long flights and inflates the apparent persistence of the MSD. The flight-level filtering therefore uses a **minimal-threshold strategy**: for each criterion, the threshold is set to the smallest value that still clears non-genuine flights (sled runs, tows, aborted or truncated logs), so the ensemble is disturbed as little as possible. Once chosen, each cut is verified by the full-distribution diagnostics in Fig. 2.3: it sits in the empty tail of its distribution, where moving it changes almost nothing. All flight-level quantities are computed directly from the parsed tracks; the thresholds follow the minimal-threshold strategy just described. Two track quantities gate a flight, plus a metadata stratifier:

- (1) *Recorded duration* $T \geq 40$ min. A cross-country flight spans many climb–glide cycles; shorter logs are sled runs or aborted attempts. An upper sanity bound $T \leq 16$ h removes loggers left running past the flight: the census holds fifteen “flights” of 16 h to 166 h, with paths up to 10,000 km, which would poison the long-lag MSD. The cut is deliberately *decoupled* from the maximum MSD lag – the ensemble at lag t already uses only flights with $T > t$, and the aging in T is diagnosed by stratifying in T – so large-lag analyses restrict to the long-duration stratum rather than raising this global cut.
- (2) *Flown path length* $L \geq 10$ km, with $L = \sum_i \Delta s_i$ from the great-circle steps of (2.1), a floor against degenerate logs. An earlier draft used 30 km; the census showed that value violating the minimal-threshold strategy itself. Above 40 min, $L < 10$ km catches 6 flights out of 184,583, while $L < 30$ km removed 5,229 genuine, localized soaring flights (median extent 7.4 km) – precisely the most-trapped tail of the ensemble, whose exclusion would bias the transport toward travellers. Duration already clears tows and sled runs.
- (3) *Altitude activity*. The total range of the adopted altitude channel must reach 75 m: a ≥ 40 min record spanning less is not free flight (a vehicle log, or a dead sensor). The census holds a few hundred such candidates among retained paraglider records; they are counted and dropped with the reason recorded.
- (4) *Glider class* (metadata). For class-stratified analyses, flights whose `aile_class` cannot be normalised onto the discipline’s system are dropped from the class-resolved sub-analyses but kept in the pooled one. Stratification by glider *type* (paraglider vs hang glider) is already possible; the sailplane stratum follows once that source is in hand.

A separate minimum-fix-count cut would be redundant: at $T \geq 40$ min even a 10 s logger – the slowest common hang-glider rate (Fig. 2.7) – already records ~ 240 fixes, ample for the kinematics; the thin tail of still slower loggers is handled by the Δt stratification (Sec. 2.7.6), not by a fix-count cut. A further criterion, sampling regularity, is applied later, where a flight’s time base is examined (Sec. 2.7.6) (implementation details: Sec. 2.6.4).

Figure 2.3 shows, for every discipline and computed on the *full* dataset (every track, no sub-sampling), the distributions of the two track cuts and, beside each, the fraction of flights that cut *alone* would retain – its marginal effect, not a cascade. Both cuts sit far out in the tail, so the retained fraction is insensitive to their exact value; the diagnostics must be re-examined for the less curated further sources.

2.7.5 From geographic to local Cartesian coordinates

The cleaned, trimmed track is next mapped from geographic to local Cartesian coordinates; the two remaining steps (resampling and smoothing) are naturally metric and run on the result. Latitudes and longitudes are angles on a curved surface and cannot be differenced directly to obtain distances; the mapping proceeds in two steps, following the convention of Vilpellet et al.

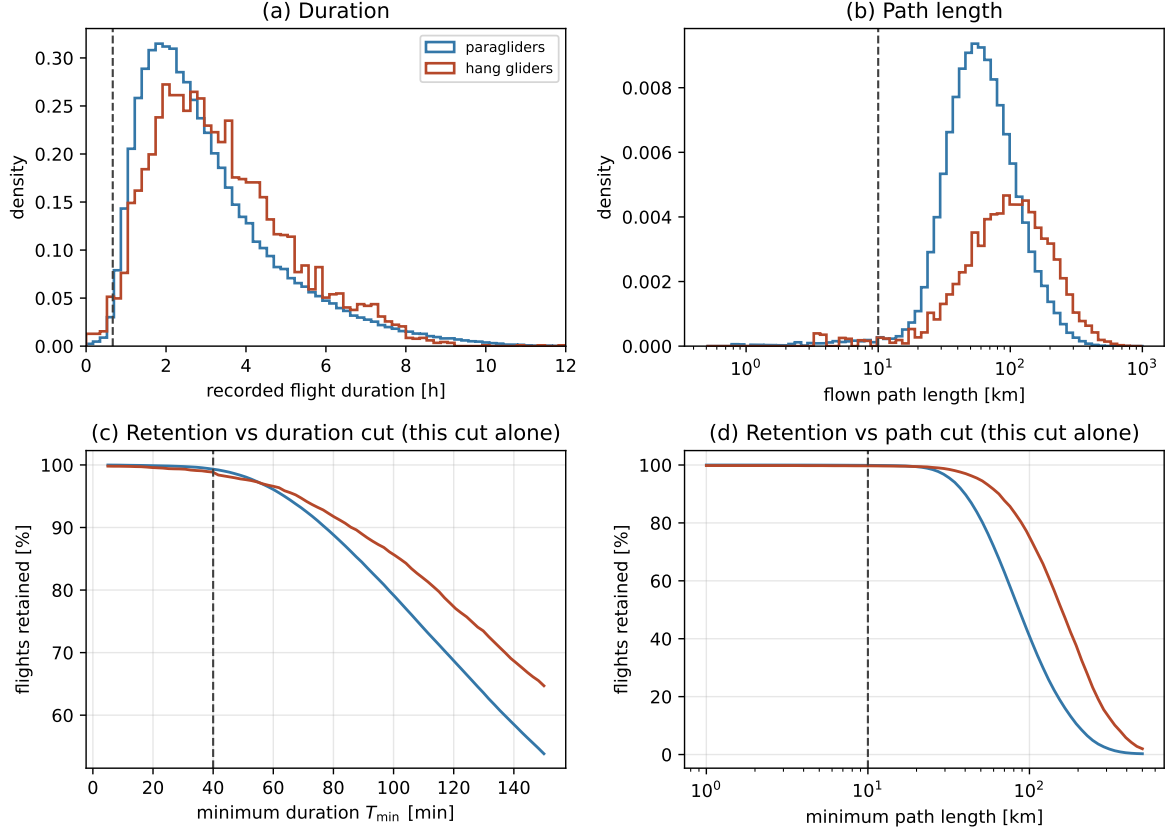


Figure 2.3: Flight-level filtering diagnostics, computed on the *full* dataset – every parsed track, per discipline (paragliders and hang gliders; sailplanes later) – not from the declared metadata and not sub-sampled. Top: distributions of (a) recorded flight duration and (b) flown path length. Bottom: the fraction of flights retained versus (c) the duration cut and (d) the path-length cut, each computed for *that cut alone* (marginal, not cascaded). Dashed lines mark the adopted cuts ($T \geq 40$ min, $L \geq 10$ km), which sit far out in the tail. (implementation details: Sec. 2.6.4)

Step 1: geodetic to ECEF. Each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6,378,137$ m, flattening $f = 1/298.257223563$, $e^2 = f(2 - f)$) [7, 8],

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \quad \begin{aligned} X &= (N + h) \cos \varphi \cos \lambda, \\ Y &= (N + h) \cos \varphi \sin \lambda, \\ Z &= (N(1 - e^2) + h) \sin \varphi, \end{aligned} \quad (2.5)$$

where $N(\varphi)$ is the prime-vertical radius of curvature and h the ellipsoidal height, measured along the ellipsoid normal.

Geodetic versus geocentric latitude. The latitude φ in (2.5) is the *geodetic* latitude – the angle between the equatorial plane and the ellipsoid *normal* at the point (Figure 2.4), *not* the geocentric latitude ψ , which is the angle to the line from the Earth’s centre O . On an ellipsoid the two differ, because the normal does not pass through O (Figure 2.4): the gap $\varphi - \psi$ reaches $\approx 0.19^\circ$ (about $11'$) near 45° , which if the two were confused would misplace a fix by up to ~ 20 km.

We use the geodetic latitude throughout, and no conversion is needed: WGS84 – hence the GPS receiver and the IGC B records (Sec. 2.2) – reports geodetic latitude, and (2.5) with

$N(\varphi)$ is precisely the geodetic-to-ECEF transform that consumes it. The longitude λ carries no such ambiguity: it is the same angle in both conventions, since the ellipsoid is a surface of revolution about the polar axis. The geodetic latitude and longitude are exactly the (φ, λ) that orient the local frame below.

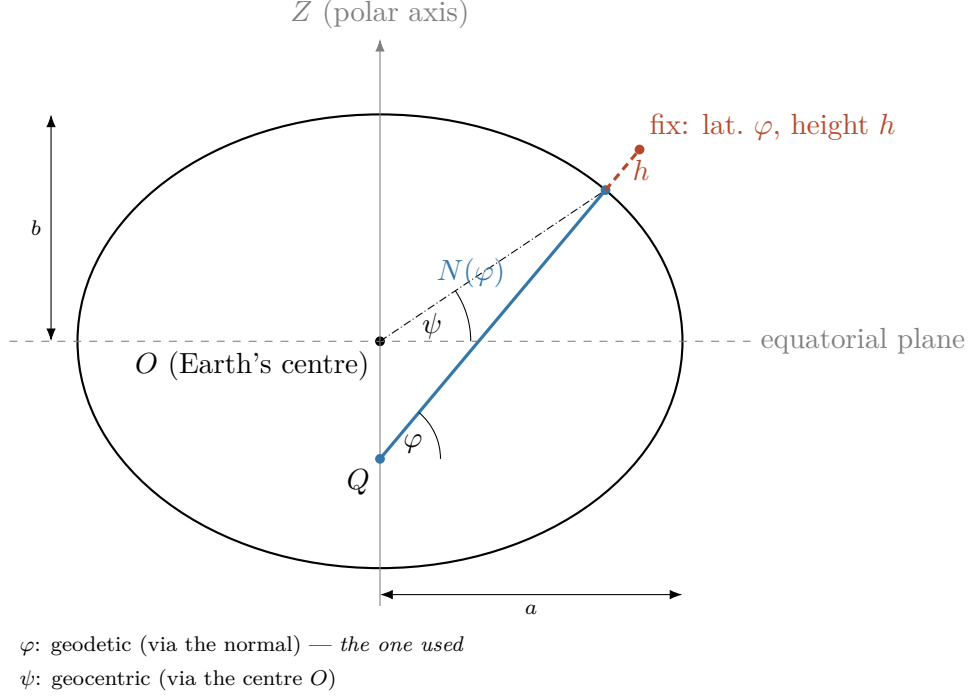


Figure 2.4: Geodetic latitude and height on the reference ellipsoid, in meridian cross-section (flattening exaggerated for visibility; true WGS84 flattening $f \approx 1/298$ would look indistinguishable from a circle at this scale): the semi-major and semi-minor axes (a, b) , the geodetic latitude φ – the angle between the ellipsoid normal and the equatorial plane – and the height h above the ellipsoid along that normal. The normal does *not* pass through the centre O except at the pole and the equator; $N(\varphi)$ is precisely the distance, along the normal, from the ellipsoid surface to the point Q where it crosses the polar axis. Because of this, the geodetic latitude φ (via the normal, the one WGS84/GPS reports and the one used here) differs from the *geocentric* latitude ψ (dash-dotted, the angle of the radius $O-P$): $\psi < \varphi$ off the pole and equator. The third geodetic coordinate, longitude λ , is not visible in this cross-section (it is the angle *around* the polar axis, out of the page here) and is shown instead in Figure 2.5, together with how (φ, λ) orient the local frame used throughout.

Step 2: ECEF to local ENU. Taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the flight – the take-off fix, since the ground phase has already been trimmed – as the origin, the ECEF displacement $\Delta \mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin \lambda_0 & \cos \lambda_0 & 0 \\ -\sin \varphi_0 \cos \lambda_0 & -\sin \varphi_0 \sin \lambda_0 & \cos \varphi_0 \\ \cos \varphi_0 \cos \lambda_0 & \cos \varphi_0 \sin \lambda_0 & \sin \varphi_0 \end{pmatrix} \Delta \mathbf{X}. \quad (2.6)$$

The ENU frame *is* the local Cartesian frame: its horizontal plane (E, N) is the “north–east” plane and U is the local vertical, so there are not two distinct coordinate systems but one, reached through the intermediate (global) ECEF representation. Over the spatial extent of a single flight (tens of kilometres) the tangent-plane approximation is accurate to well below

the GPS noise. Figure 2.5 shows the action of the rotation matrix: it turns a vector's global ECEF components into local East, North, Up ones.

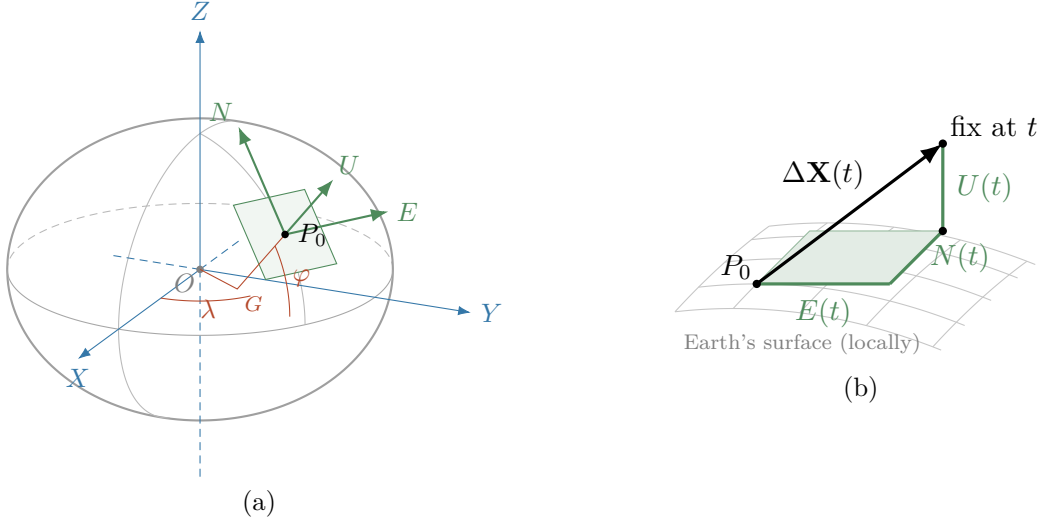


Figure 2.5: What the rotation matrix does, reproduced in our notation. (a) The global ECEF frame (X, Y, Z) in blue, centred at Earth's centre O ; the two red angles that locate the take-off fix P_0 – longitude λ , measured in the equatorial plane from the prime meridian (X), and the geodetic latitude φ , the tilt of the ellipsoid normal, measured at the point G where that normal meets the equatorial plane; and the local ENU frame (E, N, U) in green, tangent to the ellipsoid at P_0 . The globe is drawn as an oblate ellipsoid (flattening exaggerated, as in Figure 2.4), which makes visible that U follows the geodetic normal $G-P_0$ and *not* the radius from O . The green arrows are the very directions the rotation matrix sends X, Y, Z onto – its three rows, as unit vectors. (b) The same tangent plane in close-up (N receding into the page). A later fix's ECEF displacement $\Delta \mathbf{X}(t) = \mathbf{X}(t) - \mathbf{X}_0$ is the space diagonal (black) of the box whose edges are its ENU components $E(t), N(t), U(t)$ – exactly the three rows of the matrix above, applied to $\Delta \mathbf{X}$.

The altitude h that enters this transform is the adopted barometric channel (Sec. 2.7.1); changing channel would move (E, N) by a relative $h/(N + h) \sim 10^{-5}$, i.e. negligibly, so the horizontal coordinates are insensitive to the altitude source. The vertical coordinate $z = U$ is therefore, up to the small deterministic tangent-plane curvature term, just the barometric height referred to the take-off value, $z \approx h - h_0$ – exactly the quantity whose dynamics were argued above to be cleanest.

2.7.6 Uniform sampling within a flight

The smoothing and differentiation below assume a *uniform* sampling interval, so each surviving flight's time base is examined next. Loggers record at a fixed nominal rate – paragliders mostly at 1 s (69 % of flights), hang gliders spread across several rates from 1 s to 10 s with a thin tail beyond (Fig. 2.7) – but the realised series can carry jitter and drop-outs. We keep each flight at its *native* rate rather than resampling to a common cadence: a common grid would either upsample slow loggers (inventing information) or throw away the resolution of fast ones, and it is unnecessary, since the derivative operator below is given the flight's own Δt (Sec. 2.7.7). What matters is uniformity *within* a flight, not across flights.

Per flight, let Δt be the median inter-fix interval. Three cases arise:

Uniform. All intervals equal Δt to tolerance: the series is used as is.

Mildly irregular. Isolated jitter or single drop-outs – no gap beyond the split bound $g_{\max} = \min(10 \Delta t, \max(20 \text{ s}, 2 \Delta t))$ and less than 10 % of the grid missing (the sampling thresholds, Sec. 2.6.6): the flight is resampled onto the uniform grid at its native Δt , interpolating across the small gaps only, and the interpolated fraction is recorded.

Broken by a long gap. A gap longer than $g_{\max}$ – native, or opened where cleaning removed a frozen-lock run (Sec. 2.7.2) – is not bridged: interpolating across it would fabricate dynamics the smoothing could not tell from real motion. The flight is *split* at the gap into independently analysed segments, each carried forward on its own; what a split means for the analysis is stated below. A residual missing fraction above 10 % within a segment drops that segment alone, and the flight only if no segment survives.

The split bound has two scales. The relative part, $10 \Delta t$, tolerates logger hiccups in proportion to the cadence and caps the number of interpolated grid points at ten. Alone, however, it would let a slow logger bridge very long holes: 30 s at $\Delta t = 3 \text{ s}$, 100 s at $\Delta t = 10 \text{ s}$. The harm of a bridged gap is set by the motion’s own timescales, not the logger’s: an interpolated stretch is a straight line, so a gap approaching the thermalling period ($\sim 30 \text{ s}$) could replace a full circle, and the segmentation downstream would read the invented straight stretch as a glide. The absolute cap of 20 s is pinned by three constraints: at least $2\times$ the slowest common cadence (10 s, Fig. 2.7), so a single missed fix never splits a flight; at most about two thirds of a thermalling period, so a bridged gap can never hide a full circle; and well below $\tau_{\text{freeze}} \gtrsim 60 \text{ s}$ to 90 s, for consistency – an excised frozen-lock run of that length is split, so a native gap of comparable length cannot be bridged. One refinement keeps the first constraint true at *every* cadence: the cap is floored at $2 \Delta t$, i.e. $g_{\max} = \min(10 \Delta t, \max(20 \text{ s}, 2 \Delta t))$, so the thin tail of very slow loggers (Fig. 2.7) also tolerates a single missed fix; at those cadences a bridged gap inserts at most two grid points. At the dominant 1 s cadence the relative bound binds ($g_{\max} = 10 \text{ s}$); from $\Delta t = 2 \text{ s}$ to 10 s the 20 s cap does; beyond, the $2 \Delta t$ floor.

On the census, the combined bound leaves 15.3 % of paraglider and 40.5 % of hang-glider flights with at least one split, against 11.8 %/14.5 % under the relative rule alone; the increase sits at the slow cadences, where a 20 s hole is only a few missed fixes. A split is cheap for the ensemble observables (below) but fragments the within-segment statistics, so the cap’s working value is one of the parameters swept in the downstream-invariance check of Sec. 2.7.2 before being frozen.

Consequences of a split. The observables of Ch. 3 are affected differently by a split, and the distinction is worth stating once. In practice a split is bookkeeping, not new trajectories: the conversion to ENU (step v) runs before this step, so every fix already carries coordinates and elapsed time referred to the parent take-off, and the split only assigns a segment identifier. Ensemble quantities (the MSD, the propagator) need the position at a given elapsed time, not the path between: a segment keeps its parent’s origin and clock (Sec. 2.7.8), and a GPS position is absolute, so the positions after the gap remain valid and the flight simply contributes no data at the missing lags. Path-dependent quantities (path length, phase durations, step lengths, time-averaged windows) are computed within a segment and never across the gap. Three cautions follow.

- *Censoring at the boundary.* A phase in progress when the gap opens is truncated: its duration is a lower bound, not a measurement. Boundary-truncated durations are flagged as censored and excluded from the $\psi(\tau)$ fits, or handled with censoring-aware estimators, rather than counted as complete; otherwise every split would read as a spuriously short wait and thin the same tail the cleaning protects.
- *A segment is not a fresh flight.* For observables that depend on the time since take-off,

such as the time-averaged MSD stratified by duration (Ch. 3), a segment beginning at parent time $t_0 > 0$ is an observation window of an aged process, and is used as such rather than re-zeroed.

- *No second pass of the flight-level cuts.* The parent flight has already passed the cuts of Sec. 2.7.4, and re-applying the duration cut to segments would preferentially discard the data behind late-flight gaps. A segment needs only to support the smoothing window and the within-segment statistics: one below a minimal duration (a configuration key set with the implementation, Sec. 2.6.6) is dropped alone, and the rest of the flight is retained.

Interpolation across short gaps. The fill is done per channel on the uniform grid. The horizontal position, which the transport analysis measures directly, is interpolated shape-preservingly (a monotone piecewise-cubic) and only ever across the short gaps a split has left behind; the barometric altitude, a smooth, low-noise, slowly-drifting sensor within a phase (Sec. 2.7.1), is interpolated linearly (or by a cubic spline) with negligible error over such gaps. Both are filled to the *same* completeness invariant: within a retained segment every grid time carries a defined (x, y, z) , so the segmentation downstream is never handed a missing value. Every filled point carries an **interpolated** flag: the invariant serves the segmentation, but downstream statistics can distinguish measured from reconstructed samples, and a phase boundary that falls inside a bridged stretch inherits the same caution as one at a segment boundary. The altitudes deferred in Sec. 2.7.2 (a dropped barometric spike, a V fix on a GNSS-fallback flight) are restored here.

One might expect the altitude to tolerate wider gaps, since the vertical speed is about an order of magnitude smaller than the horizontal ($v_z \sim 1 \text{ m s}^{-1}$ against $v_{xy} \sim 10 \text{ m s}^{-1}$). It does not, and the gap bound is the same for every channel: the interpolation error is set by the *curvature* of the signal over the gap, not by its speed (a fast but straight motion is interpolated exactly, a slow but turning one is not), so a slower channel is not automatically a smoother one.

More decisively, the altitude feeds v_z , the segmentation’s discriminant, and a long altitude gap linearly interpolated returns a single average slope that *erases any climb–glide transition falling inside it*; and those transitions are exactly where v_z swings fastest (a thermal entry flips v_z from about -1.5 to $+3 \text{ m s}^{-1}$ in a second or two), the worst case for interpolation. One asymmetry is real and already used: an *isolated* missing altitude is filled far more reliably than a horizontal one, because the barometric channel is so smooth. This licenses “drop the altitude only” at cleaning, but it says nothing about widening the gap.

The gap bound is the sampling-regularity control anticipated in Sec. 2.7.4; Figure 2.6 makes the fraction of flights it leaves whole auditable in the same way as the duration and path-length cuts (implementation details: Sec. 2.6.6). One caveat carries downstream: a few observables are intrinsically Δt -dependent (a finite-difference velocity, a turning angle), so where they are compared across flights the ensemble is stratified by rate or restricted to the dominant one, rather than pretending a single cadence.

2.7.7 Smoothing and differentiation

Velocity and acceleration drive the kinematics and the segmentation, but differencing raw GPS positions amplifies the high-frequency noise characterised in Sec. 2.7.1. A **Savitzky–Golay** filter [9] solves both problems at once: it fits a low-order polynomial to a sliding window by least squares and reads off that polynomial, or its derivatives, at the window centre, returning smoothed position, velocity and acceleration (deriv = 0, 1, 2) in one pass per component. The derivative outputs are scaled by the flight’s own Δt (Sec. 2.7.6), so the multi-rate fleet needs

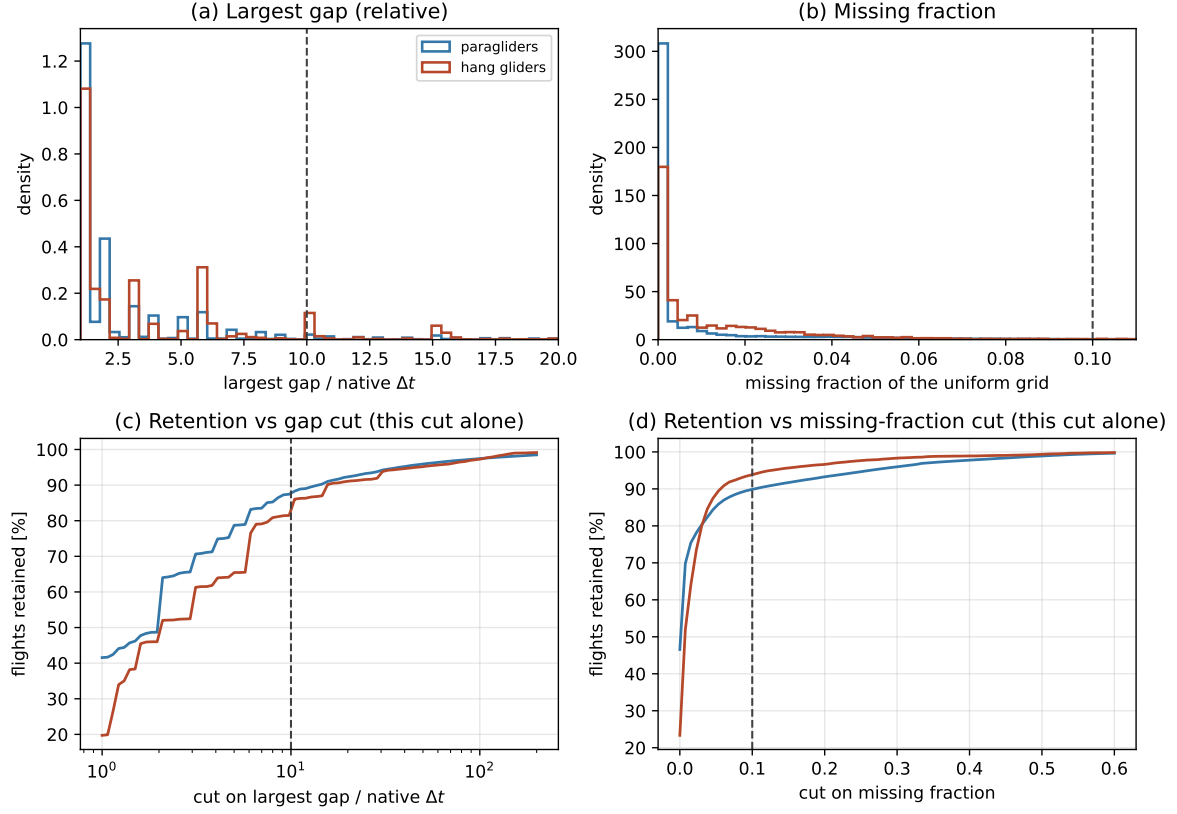


Figure 2.6: Sampling-regularity diagnostics, computed on the full dataset per discipline, on the *same* track scan as Figure 2.3 (so every parsed flight is included, not only the ones that survive the duration/path-length cuts). (a)/(b) Distribution of the largest single gap (relative to that flight’s own native Δt) and of the missing fraction of a uniform grid at that interval, both on linear bins and a linear axis, over the bulk of the distribution (the heavy tail is not lost: (c)/(d) sweep the full range). Unlike native Δt (Figure 2.7) the largest-gap ratio is not purely discrete (a missed fix or two at the native rate is the most common cause of a large gap, but the gap’s exact duration is not grid-aligned to it), yet integer ratios (a gap of exactly k native intervals) are still visibly the dominant, common values, especially for hang gliders; a logarithmic axis, appropriate for a quantity spanning orders of magnitude, would have stretched precisely this small-integer structure into invisibility; the structure matters here because the adopted cut (dashed) sits on it. (c)/(d) The fraction of flights each cut would retain *on its own* (marginal, not cascaded), with the adopted thresholds marked; here the cut ranges over orders of magnitude to show the full saturating shape, so (c) keeps a logarithmic axis. (implementation details: Sec. 2.6.6)

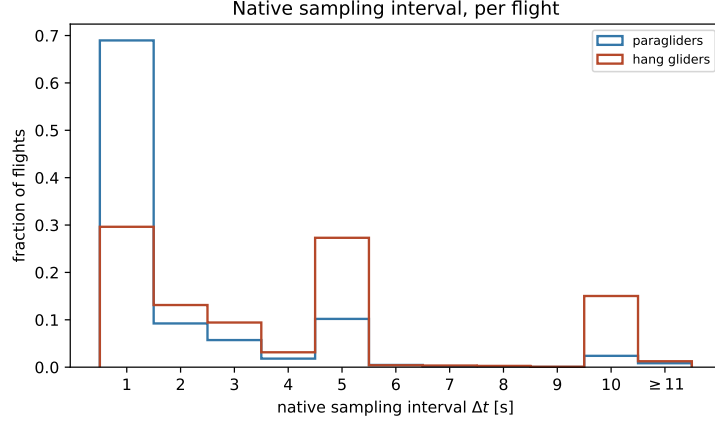


Figure 2.7: Native sampling interval Δt (median inter-fix time) per flight, on the full dataset per discipline. Δt is, in practice, a *discrete* quantity, not a continuous one: it lands on an exact whole second for the large majority of flights in both disciplines (a logger reports at one of a handful of fixed configured rates), so bins are one second wide and centred on each integer – each bar is the fraction of flights at exactly that rate, not a density (with unit-width bins the two coincide numerically, but only “fraction” describes what is actually being counted). The fleet is multi-rate: paragliders concentrate heavily at 1 s, whereas hang gliders spread across several rates, with a long thin tail beyond 11 s (last bin, aggregated). A single common cadence would waste the fast loggers; instead each flight keeps its own Δt , which the smoothing step consumes directly (Sec. 2.7.7). (implementation details: Sec. 2.6.6)

no common grid, and the window never crosses a segment boundary. Because it runs on the resampled grid (Sec. 2.7.6), the filter sees a uniform time base with the short gaps already filled; the interpolated points (a small, recorded fraction, since heavily reconstructed flights are dropped whole, Sec. 2.7.2) add a little local correlation but no bias over gaps this short.

The filter has two hyperparameters, the window length w (odd, in samples) and the polynomial order p , fixed from the measured noise spectrum rather than by eye:

- (i) From the ensemble Welch power spectrum of the raw ENU components on a sample of flights (the diagnostic of Sec. 2.7.1 applied to E , N , U), read the frequency f_c of the knee where the dynamical signal meets the flat GPS noise floor. Its reciprocal is a smoothing timescale $\tau_c = 1/f_c$: dynamics slower than τ_c are signal, faster fluctuations are noise.
- (ii) Fix p on a physical argument: to represent local curvature and give a clean second derivative the fitted polynomial must be at least quadratic, while a low order keeps the pass-band flat; $p = 3$ works and the result is insensitive to p within $\{2, 3, 4\}$.
- (iii) Set the window per flight from the *timescale*, $w = \text{odd}(\tau_c/\Delta t)$, so the physical smoothing scale is the same across the fleet regardless of rate – with a floor at $w = 5$, the smallest window a cubic fit accepts. For the few flights with $\Delta t \gtrsim \tau_c$ the noise band lies beyond Nyquist and the floor smooths slightly more than the noise scale; those flights are in any case stratified out of the Δt -sensitive observables (Sec. 2.7.6).
- (iv) Treat the vertical separately from the horizontal, and condition it on the altitude *source*. On a barometric flight the vertical channel is *smoother* than the horizontal, since the pressure sensor carries little high-frequency noise (Sec. 2.7.1), so it takes a shorter window; on a GNSS-fallback flight the vertical is instead $1.5\text{--}3\times$ noisier (vertical dilution of precision) and needs more smoothing. Its knee f_c , and hence its window, are therefore read from the spectrum of the channel actually in use, separately from those of (E, N) .

The choice is validated on a held-out sample: the residual (raw minus smoothed) spectrum should be flat above f_c , confirming that noise rather than signal was removed; the acceleration distribution should stay within physical bounds; and the results should be stable under a one-step change of w (implementation details: Sec. 2.6.7).

2.7.8 Notation

Throughout, $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$ denotes the *horizontal* position in this local frame and $z(t) = U(t)$ the (barometric) altitude. By construction each trajectory starts at the horizontal origin, $\mathbf{r}(0) = \mathbf{0}$, with the clock reset to $t = 0$ at take-off, so t is the elapsed flight time (a segment created by a split at a long gap, Sec. 2.7.6, keeps its parent flight’s origin and clock). The vertical is *not* re-zeroed: the take-off height is retained, so the analysis keeps both the increment that drives the dynamics (invariant to any offset) and the absolute altitude for interpretation (potential energy, convective-boundary-layer structure). The horizontal velocity is $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$, with magnitude the horizontal speed $v_{xy} = |\mathbf{v}|$; the vertical velocity is $v_z = \dot{z}$. The horizontal displacement over a lag t is $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$.

2.8 Preliminary characterization

Before the main transport analysis it is useful to run a set of lightweight checks that validate basic assumptions, reveal potential stratification needs, and characterize the dataset at the level of individual trajectories. These do not require segmentation and are conceptually simpler than the transport observables, but they are prerequisites for a well-founded analysis.

Trajectory statistics. For each flight, computed from its parsed track: airborne duration T , total horizontal path length, number of valid fixes and native sampling interval Δt . Their ensemble distributions (the filtering-relevant ones in Fig. 2.3) characterise what the dataset contains, flag outliers, and, through Δt , set the per-flight smoothing window (Sec. 2.7.7). In the present data durations centre on a few hours and track lengths on $\sim 10^4$ fixes, so the flight-level cuts fall in an essentially empty tail.

Isotropy check. The scaling of the 1D marginal propagator (Section 3.1) requires the process to be isotropic. Competition flights have preferred directions (race goal, orography, prevailing winds), so isotropy must be tested rather than assumed.

The primary test is the **per-component variance ratio**: for an isotropic process $\langle E(t)^2 \rangle = \langle N(t)^2 \rangle = \text{MSD}(t)/2$ for all t , so the ratio $\langle E(t)^2 \rangle / \langle N(t)^2 \rangle$ should be identically 1. Plotting it as a function of t reveals both the degree and the scale-dependence of any anisotropy. A complementary check is a Kolmogorov–Smirnov test comparing the empirical distributions of $E(t)$ and $N(t)$ at a few selected lags.

If isotropy fails, the two marginals $P(E(t) = x)$ and $P(N(t) = x)$ are analysed *separately*: this is both the natural fallback and informative in itself, since unequal scaling exponents $H_E \neq H_N$, or equal exponents with different scaling functions, signal geometrically anisotropic transport.

Compatibility across strata. The dataset spans more than two decades, two glider types (paragliders and hang gliders), and a range of performance classes (Sec. 2.4), from recreational EN A/B paraglider wings to CCC racing wings and the flexible/rigid hang-glider classes, whose kinematics differ substantially; sailplanes will later add a third type. Before treating all flights as draws from a single statistical process, i.e. before computing ensemble averages over the pooled dataset, the subpopulations must be checked for statistical compatibility.

A lightweight way to do this: for each stratum (e.g. a single wing class, or the flights of a single season) compute the per-component displacement variance $\langle E(t)^2 \rangle$ as a function of the elapsed time t and overlay the curves on one plot. If they coincide, the transport statistics are compatible and pooling is justified. If one stratum grows systematically faster or with a different shape (competition wings, say, cover ground faster than recreational ones), the strata must be analysed separately, and that difference is itself a result about the universality of the transport regime.

Spatial coverage. A map of take-off coordinates and displacement end-points at fixed t reveals geographic clustering (France has distinct orographic zones: Alps, Pyrenees, Massif Central, plains). If flights from mountainous terrain differ markedly from flatland flights, geographic stratification may be needed alongside temporal and aeronautical stratification.

Chapter 3

Next steps

With the dataset assembled, cleaned and characterised (Chapter 2), the analysis proceeds as follows.

3.1 Observables on the whole trajectory

A first group of observables requires *no* segmentation: it can be measured directly on the full trajectories and already probes the transport at the global level. For each, we state what it is, how it is computed, and what it is for.

Transport and diffusion.

Mean-squared displacement (MSD). With every flight starting at the origin ($\mathbf{r}(0) = \mathbf{0}$), the *ensemble-averaged* MSD is the mean squared horizontal distance reached after an elapsed time t ,

$$\text{MSD}(t) = \langle |\mathbf{r}(t)|^2 \rangle, \quad (3.1)$$

where $\langle \cdot \rangle$ is an average *over flights* (an ensemble average) at fixed t . Its growth law $\text{MSD}(t) \sim t^\alpha$ fixes the transport regime ($\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$ sub-diffusive; equivalently the Hurst exponent $H = \alpha/2$). This is the primary test for anomalous transport.

Time-averaged MSD (TA-MSD). A second MSD estimator is built entirely *within* a single flight. For flight k of duration T_k , one slides a window of width t along the trajectory $\mathbf{r}^{(k)}$ and averages the squared displacement over all starting times t_0 *inside that flight*:

$$\overline{\delta^2}^{(k)}(t) \equiv \frac{1}{T_k - t} \int_0^{T_k - t} |\mathbf{r}^{(k)}(t_0 + t) - \mathbf{r}^{(k)}(t_0)|^2 dt_0. \quad (3.2)$$

The average over t_0 here is a *time* average internal to flight k ; it has nothing to do with the ensemble average $\langle \dots \rangle$ over different flights used in the MSD. Because a single-flight curve $\overline{\delta^2}^{(k)}(t)$ is noisy, the operationally useful quantity is its mean over all flights of comparable duration T :

$$\text{TAMSD}(t, T) \equiv \left\langle \overline{\delta^2}^{(k)}(t) \right\rangle_{T_k \approx T}. \quad (3.3)$$

Ergodic case. For an ergodic process, $\text{TAMSD}(t, T) \rightarrow \text{MSD}(t)$ as $T \rightarrow \infty$, independently of T , and the individual values $\overline{\delta^2}^{(k)}(t)$ concentrate around the common limit (the distribution collapses to a delta function).

Non-ergodic case: two independent signatures [10].

- (i) *Aging.* TAMSD(t, T) differs from MSD(t) and retains a systematic dependence on T : flights of different duration yield different curves. This is a strong indicator of non-ergodicity, but it is sufficient, not necessary – some non-ergodic processes show only a weak T -dependence.
- (ii) *Non-self-averaging.* The distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories remains broad even as $T \rightarrow \infty$: individual flights give systematically different TA-MSD values and the distribution does not collapse. This is the more fundamental signature: it persists even when aging is weak.

We therefore monitor both TAMSD(t, T) and the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories at fixed t and T . The canonical illustration – a CTRW with power-law waiting times – is worked out analytically in Appendix A, where both EA-MSD $\neq$ TA-MSD and the T -dependence are derived explicitly.

Propagator. Since each flight starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, the position at time t and the displacement from the start coincide: $\mathbf{r}(t) = \mathbf{r}(t) - \mathbf{r}(0)$. We therefore define the propagator as the probability density of the position vector $\mathbf{x} \in \mathbb{R}^2$ at time t ,

$$P(\mathbf{x}, t) d^2x = \Pr(\mathbf{r}(t) \in [\mathbf{x}, \mathbf{x} + d\mathbf{x}]). \quad (3.4)$$

For a *self-similar* process with exponent H the propagator obeys the scaling form

$$P(\mathbf{x}, t) = t^{-2H} G\left(\frac{\mathbf{x}}{t^H}\right), \quad (3.5)$$

where $G : \mathbb{R}^2 \rightarrow \mathbb{R}_{>0}$ is a time-independent scaling function (normalised, $\int G d^2u = 1$).

In practice one works with the **one-dimensional marginal** obtained by projecting onto a single Cartesian component x ,

$$P(x, t) \equiv \int_{-\infty}^{+\infty} P((x, y), t) dy. \quad (3.6)$$

Assuming the process is *isotropic*, $G(\mathbf{u}) = G(|\mathbf{u}|)$, the integral can be evaluated by the substitution $y = t^H v$:

$$P(x, t) = t^{-2H} \int_{-\infty}^{+\infty} G\left(\sqrt{\left(\frac{x}{t^H}\right)^2 + v^2}\right) t^H dv = t^{-H} F\left(\frac{x}{t^H}\right), \quad (3.7)$$

where $F(u) \equiv \int_{-\infty}^{+\infty} G(\sqrt{u^2 + v^2}) dv$ inherits the normalisation $\int F du = 1$ from G . The marginal therefore satisfies the same scaling form as the 2D propagator, with the same exponent H but in one dimension. The shape of F distinguishes transport regimes: a Gaussian F characterises ordinary diffusion, while heavy power-law or stretched-exponential tails signal anomalous transport.

The marginal $P(x, t)$ is the object actually estimated from data: use the East component, the North component, or both (if isotropy holds, they coincide and can be pooled; if not, they are analysed separately – see Section 2.8). The scaling collapse consists in plotting $t^H P(x, t)$ against x/t^H for several values of t : all curves should fall on the single function F .

Probability of return to the origin. The value of the propagator at its centre, $P_0(t) \equiv P(\mathbf{0}, t)$, i.e. the height of the peak. For a self-similar process it decays as a power law set by the same exponent, $P_0(t) \sim t^{-dH}$ in d dimensions (t^{-2H} in the plane, t^{-H} for a

single component). It therefore yields H from the *bulk* of the distribution rather than from its tails, which is valuable when heavy tails make the variance – and hence the MSD – ill-defined; a change of slope in $\log P_0$ versus $\log t$ exposes a crossover between regimes, and a peak exponent that disagrees with the MSD exponent is itself informative, pointing to multi-scaling. (This is the route Mantegna and Stanley [11] used to read off the index of Lévy-like walks.)

Non-Gaussianity. How far the propagator departs from a Gaussian, measured by the non-Gaussian parameter (in the plane)

$$\alpha_2(t) = \frac{\langle |\mathbf{r}(t)|^4 \rangle}{2 \langle |\mathbf{r}(t)|^2 \rangle^2} - 1, \quad (3.8)$$

which vanishes for a Gaussian and is positive for heavier-than-Gaussian tails (the factor 2 is the two-dimensional Gaussian value of $\langle |\mathbf{r}|^4 \rangle / \langle |\mathbf{r}|^2 \rangle^2$). Evaluated on the propagator $P(\mathbf{x}, t)$ at each elapsed time t , it is a function of t (one number per time) tracing a curve $\alpha_2(t)$ that pinpoints the scales over which the displacement statistics are non-Gaussian (typically largest at intermediate times, where occasional long glides dominate, and decaying if a central-limit Gaussianisation sets in at long times). It summarises the shape of P without replacing it.

Velocity autocorrelation $C(\tau)$. $C(\tau) = \langle \mathbf{v}(t_0) \cdot \mathbf{v}(t_0 + \tau) \rangle$ (often normalised by $C(0)$) is the dot product of the velocity with its later self: it is large and positive when, a delay τ later, the motion still points the same way, so the scale over which $C(\tau)$ decays sets the memory time of the heading. It controls the MSD through the Green–Kubo relation (for a stationary velocity) $\text{MSD}(t) = 2 \int_0^t (t - \tau) C(\tau) d\tau$: a slowly decaying C produces a faster-growing MSD. The functional form of $C(\tau)$ is the diagnostic: an integrable C relaxes to normal diffusion, a non-integrable (e.g. power-law) tail sustains persistent super-diffusion with no crossover, and a sign change reveals the anti-persistence expected from circling.

Turning-angle distribution. The distribution of the angle θ between two successive horizontal displacement vectors along the trajectory – the change of heading from one step to the next. A distribution peaked at $\theta = 0$ means strong forward persistence (a correlated random walk); a flat distribution means memoryless reorientation. This is the microscopic origin of the persistence seen in $C(\tau)$ and the MSD.

First-passage and first-exit times. The first-passage time $T(L)$ is the first time the horizontal distance from take-off reaches a value L (the first-exit time, the first time the trajectory leaves a disc of radius L) [12]. For a self-similar transport process the characteristic time scales with the length as $\langle T(L) \rangle \sim L^{1/H}$, with dynamic exponent $z = 1/H$, so the slope of $\log \langle T \rangle$ against $\log L$ gives an estimate of H obtained in the space domain, independent of the MSD; the full distribution of $T(L)$ additionally exposes heavy tails coming from long trapping events.

Speed and vertical-velocity distributions. The magnitude of the horizontal velocity and the vertical velocity. Basic kinematic characterization; the vertical velocity also separates climbing from sinking and feeds the segmentation below.

Trajectory geometry.

Tortuosity. How much a path winds, quantified by the ratio between the length actually travelled and the straight-line distance between its endpoints over the same interval (1 for

a straight line, larger for a more convoluted path). It is a geometric *efficiency* descriptor and does not map one-to-one onto a transport exponent – very different dynamics can share a tortuosity – so we use it descriptively and to separate phases (close to 1 in glides, large in search and climb) rather than as an estimator of H .

Fractal dimension. The box-counting dimension d_f of the trail, i.e. the exponent with which the number of boxes of size ϵ needed to cover the path grows as $\epsilon \rightarrow 0$. For a self-similar walk it is tied to the transport exponent by

$$d_f = \min(1/H, 2), \quad (3.9)$$

so a ballistic line ($H = 1$) gives $d_f = 1$, a Brownian path ($H = 1/2$) gives $d_f = 2$, and the reported sub-ballistic soaring ($H \approx 0.88$) gives $d_f \approx 1.14$. The fractal dimension is thus the geometric counterpart of H and a third, independent route to it, after the MSD slope and the return/first-passage scalings; agreement among the three is a strong internal check, disagreement is evidence of multi-scaling.

Radius of gyration. The root-mean-square distance of the trajectory points from their centroid, i.e. the characteristic spatial extent of a flight. As a single number it sets the natural length scale of a flight (useful to normalise other observables); measured as it grows with elapsed time, $R_g(t) \sim t^H$, it provides a further estimate of the same exponent.

Per-flight scalars and scale dependence. The geometric quantities above can be read in two complementary ways: as a single number per flight (one tortuosity, one d_f , one R_g), whose distribution over the ensemble characterises the population and feeds the stratified comparisons; or as a function of scale or elapsed time ($R_g(t)$ as the trajectory unfolds, tortuosity on sub-segments of growing duration, the box count as a function of box size), which is where the transport exponents appear ($R_g(t) \sim t^H$; the box-count slope is d_f). The scalar is descriptive; the scale dependence links the geometry to the anomalous exponent. Both views will be used.

3.2 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [4], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed “step”.

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermallng). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized “trap”.

That work performs the segmentation with a hidden Markov model [13] whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

3.3 Phase-resolved observables

The segmentation unlocks a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 3.1, cannot be obtained without first segmenting the flight.

The waiting phase: climb and search. Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration τ_c , the search duration τ_s , and their sum $\tau_w = \tau_s + \tau_c$, the total time spent essentially stationary between glides. The distinction matters: Vilpellet et al. [4] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time $\psi(\tau_w)$, the quantity that determines whether trapping drives sub-diffusion, would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report $\psi(\tau_w)$ together with the separate $\psi(\tau_c)$ and $\psi(\tau_s)$. Alongside the durations we record the climb kinematics (altitude gained Δz , mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

The directed step: glide (transition). Each glide contributes a horizontal step of length ℓ and duration τ_g , with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic, $\ell \approx v_g \tau_g$, step length and duration are expected to be nearly proportional; we test this directly through the joint law of (ℓ, τ_g) – the ℓ – τ_g scatter, the conditional mean $\langle \ell | \tau_g \rangle$, and the spread of the ratio ℓ/τ_g (the glide speed). Both marginals matter: the tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions, so we compute the distributions of ℓ and τ_g separately as well as their joint law.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 3.1. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether ℓ and τ_w are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence*. A flight is a discrete sequence over the three states {transition, search, climb}, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix $P_{ij} = \text{Pr}(\text{next} = j \mid \text{current} = i)$: a perfectly cyclic flight ($T \rightarrow S \rightarrow C \rightarrow T$) gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide.

Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. These measurements determine whether a memoryless model is admissible.

3.4 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [4]. That study covers paragliders, hang gliders and sailplanes; the present data cover the first two (Chapter 2), so both the paraglider and the hang-glider comparisons are within reach, while the sailplane figures become reachable once that source is added. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the ~ 30 s thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios (≈ 8.5 , 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx 60\%$ transition, $\approx 35\%$ climb, the remainder searching) and the global sub-ballistic scaling ($H \approx 0.88$).

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

3.5 Transport analysis

Using these observables, test for anomalous transport: estimate the MSD scaling exponent α , fit the tails of $\psi(\tau)$ and $\lambda(\ell)$ with principled methods (e.g. maximum-likelihood power-law fits with goodness-of-fit tests [14]), and perform model selection between the continuous-time random walk and the Lévy-walk frameworks [1,2] – using the step/waiting-time coupling to tell them apart. Because logging devices, sites, gliders and weather differ, results will be stratified (by season, site, glider class, and data source) and trajectories normalized, and finite-size, censoring and sampling biases will be controlled throughout.

3.6 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of

the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

3.7 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

Bibliography

- [1] Ralf Metzler and Joseph Klafter. The random walk’s guide to anomalous diffusion: a fractional dynamics approach. *Physics Reports*, 339(1):1–77, 2000.
- [2] V. Zaburdaev, S. Denisov, and J. Klafter. Lévy walks. *Reviews of Modern Physics*, 87(2):483–530, 2015.
- [3] Gautam Reddy, Antonio Celani, Terrence J. Sejnowski, and Massimo Vergassola. Learning to soar in turbulent environments. *Proceedings of the National Academy of Sciences*, 113(33):E4877–E4884, 2016.
- [4] Jérémie Vilpelle, Alexandre Darmon, and Michael Benzaquen. From random walks to thermal rides: Universal anomalous transport in soaring flights. *arXiv:2601.01293 [cond-mat.stat-mech]*, 2026.
- [5] Fédération Aéronautique Internationale, International Gliding Commission. Technical specification for IGC-approved GNSS flight recorders. Second Edition, with Amendment 8, 2023. Defines the IGC flight-data file format. https://www.fai.org/sites/default/files/igc_fr_specification_with_al8_2023-2-1_0.pdf.
- [6] Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974.
- [7] National Geospatial-Intelligence Agency. Department of defense world geodetic system 1984: Its definition and relationships with local geodetic systems. Technical Report NGA.STND.0036, Version 1.0.0, National Geospatial-Intelligence Agency (NGA), 2014. <https://earth-info.nga.mil/php/download.php?file=coord-wgs84>.
- [8] European Space Agency. Ellipsoidal and Cartesian coordinates conversion. Navipedia. https://gssc.esa.int/navipedia/index.php/Ellipsoidal_and_Cartesian_Coordinates_Conversion.
- [9] Abraham Savitzky and M. J. E. Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639, 1964.
- [10] Y. He, S. Burov, R. Metzler, and E. Barkai. Random time-scale invariant diffusion and transport coefficients. *Physical Review Letters*, 101(5):058101, 2008.
- [11] Rosario N. Mantegna and H. Eugene Stanley. Scaling behaviour in the dynamics of an economic index. *Nature*, 376(6535):46–49, 1995.
- [12] Sidney Redner. *A Guide to First-Passage Processes*. Cambridge University Press, Cambridge, 2001.
- [13] L. R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

- [14] Aaron Clauset, Cosma Rohilla Shalizi, and M. E. J. Newman. Power-law distributions in empirical data. *SIAM Review*, 51(4):661–703, 2009.
- [15] P. Welch. The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. *IEEE Transactions on Audio and Electroacoustics*, 15(2):70–73, 1967.
- [16] Peter J. Rousseeuw and Christophe Croux. Alternatives to the median absolute deviation. *Journal of the American Statistical Association*, 88(424):1273–1283, 1993.

Appendix A

CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time T .

Model

A walker takes steps of length l_i (iid, symmetric, $\langle l^2 \rangle = \sigma^2 < \infty$) separated by waiting times τ_i (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha \tau_0^\alpha}{\tau^{1+\alpha}}, \quad \tau \rightarrow \infty, \quad 0 < \alpha < 1. \quad (\text{A.1})$$

The mean waiting time diverges ($\langle \tau \rangle = \infty$), so the walker is anomalously slow. Let $n(t)$ be the number of steps completed by time t , $T_n = \sum_{i=1}^n \tau_i$ the time of the n -th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \quad (\text{A.2})$$

the position at time t .

Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t) \rangle = \sigma^2 \langle n(t) \rangle. \quad (\text{A.3})$$

For large t the mean number of renewals in $[0, t]$ follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of ψ . Because $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau} \psi(\tau) d\tau \simeq 1 - (\tau_0 s)^\alpha + \dots$ for $s \rightarrow 0$ (a consequence of ψ having a power-law tail with exponent $1 + \alpha$), the renewal theorem gives

$$\langle n(t) \rangle \sim \frac{t^\alpha}{\tau_0^\alpha \Gamma(1 + \alpha)}, \quad t \rightarrow \infty. \quad (\text{A.4})$$

Hence the **EA-MSD** grows sub-linearly,

$$\text{MSD}(t) = \langle x^2(t) \rangle \sim \frac{\sigma^2}{\tau_0^\alpha \Gamma(1 + \alpha)} t^\alpha \equiv K_\alpha t^\alpha, \quad 0 < \alpha < 1. \quad (\text{A.5})$$

The transport is sub-diffusive with Hurst exponent $H = \alpha/2 < 1/2$.

Time-averaged MSD and its mean

For a trajectory of duration T , the time-averaged MSD at lag $t \ll T$ is

$$\overline{\delta^2}(t; T) = \frac{1}{T-t} \int_0^{T-t} [x(t_0+t) - x(t_0)]^2 dt_0. \quad (\text{A.6})$$

Since $\langle [x(t_0+t) - x(t_0)]^2 \rangle = \sigma^2 \langle n(t_0+t) - n(t_0) \rangle$, the mean of the integrand equals σ^2 times the mean number of steps in any window of width t . The key observation is that the integral $\int_0^{T-t} [n(t_0+t) - n(t_0)] dt_0$ counts, for each step k that lands at time $T_k \in [0, T]$, the length of the t -wide sliding window that contains T_k . For T_k deep in the interior ($t \leq T_k \leq T-t$) this length is exactly t ; the contributions from the two boundary layers of width t are negligible for $t \ll T$. Therefore

$$\int_0^{T-t} [n(t_0+t) - n(t_0)] dt_0 \approx t n(T), \quad t \ll T, \quad (\text{A.7})$$

and

$$\overline{\delta^2}(t; T) \approx \frac{\sigma^2 t n(T)}{T}. \quad (\text{A.8})$$

Taking the ensemble average and using $\langle n(T) \rangle \sim K_\alpha T^\alpha / \sigma^2$,

$$\text{TAMSD}(t, T) \equiv \langle \overline{\delta^2}(t; T) \rangle \approx \sigma^2 \frac{\langle n(T) \rangle}{T} t \sim K_\alpha T^{\alpha-1} t. \quad (\text{A.9})$$

Comparing (A.5) and (A.9):

- The EA-MSD scales as t^α (sub-diffusive), whereas the TAMSD scales as t^1 (linear in the lag) – a qualitatively different dependence on t .
- The TAMSD carries an explicit $T^{\alpha-1}$ prefactor: since $\alpha < 1$ this is a decreasing function of T , so longer flights yield smaller TAMSD at the same lag. This T -dependence is the signature of *aging*: the process has not forgotten its past, and the answer depends on how long we have observed it.
- The ratio $\text{TAMSD}(t, T)/\text{MSD}(t) \sim (t/T)^{1-\alpha} \rightarrow 0$ as $T \rightarrow \infty$ at fixed t – the two observables diverge.

Distribution of the TA-MSD: non-self-averaging

Equation (A.8) shows that $\overline{\delta^2}(t; T) \approx \sigma^2 t n(T)/T$, so the randomness in the TA-MSD comes entirely from $n(T)$, the total number of steps in $[0, T]$. For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T) \rangle} \quad (\text{A.10})$$

converges in distribution as $T \rightarrow \infty$ to a non-degenerate limit law: the *Mittag-Leffler distribution* M_α , whose probability density function is defined through its moments $\langle \xi^n \rangle = n! \Gamma(1+\alpha)^n / \Gamma(1+n\alpha)$ [10]. This is a broad, non-Gaussian distribution on $(0, \infty)$ that does *not* concentrate around $\xi = 1$ for $\alpha < 1$: different trajectories give systematically different values of $\overline{\delta^2}(t; T)$ even in the limit $T \rightarrow \infty$. This is *non-self-averaging*: the TA-MSD of a single trajectory is not a reliable estimate of any ensemble quantity, regardless of how long the trajectory is.

Quantitatively, the ergodicity-breaking parameter

$$\text{EB}(t, T) \equiv \frac{\langle [\overline{\delta^2}(t; T)]^2 \rangle - \langle \overline{\delta^2}(t; T) \rangle^2}{\langle \overline{\delta^2}(t; T) \rangle^2} = \text{Var}(\xi) \sim c_\alpha T^{-\alpha} \rightarrow 0 \quad (\text{A.11})$$

does tend to zero (the variance of ξ decreases with T), but the *distribution* of ξ converges to a non-trivial limit rather than to a delta function. This is why monitoring the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories – not just its mean – is the proper diagnostic for non-ergodicity: the collapse (or non-collapse) of that distribution as T grows tells us whether we are dealing with a self-averaging ergodic process or a non-self-averaging non-ergodic one.

For comparison, in an ordinary random walk ($\alpha = 1$, finite mean waiting time) the ratio $n(T)/\langle n(T) \rangle$ tends to 1 in probability as $T \rightarrow \infty$, so the distribution of the TA-MSD collapses to a delta function, $\overline{\delta^2}(t; T) \rightarrow \text{MSD}(t)$ almost surely, and the process is ergodic.

Appendix B

Power spectral density and Welch's method

This appendix explains the tool used in Chapter 2 to compare the two altitude channels of an IGC file, namely the *power spectral density* (PSD) estimated with *Welch's method* [15]. It is written so the reader need not already know what these are.

What the power spectral density is

Take a signal sampled at a fixed rate, here an altitude series $z_n = z(n \Delta t)$, $n = 0, \dots, N - 1$, with sampling interval Δt (so the sampling frequency is $f_s = 1/\Delta t$). Any such finite signal can be written as a sum of sinusoids of different frequencies (its discrete Fourier transform). The **power spectral density** describes how the power (the variance) of the signal is distributed across frequencies. A slow drift puts its power at low frequency; rapid jitter puts its power at high frequency. Formally the PSD $S(f)$ is the Fourier transform of the autocovariance of the signal, and it is normalised so that the total area under it equals the variance of the signal,

$$\int_0^{f_s/2} S(f) \, df = \text{Var}(z), \quad (\text{B.1})$$

with units of (metres)² per hertz. The upper limit $f_s/2$ is the *Nyquist frequency*: a series sampled every Δt cannot resolve anything faster than one oscillation per two samples. A PSD thus shows where in frequency the signal's power lies, which is what separates a slowly drifting but locally clean channel (power concentrated at low f) from a channel with a high-frequency noise floor (power that stays large up to the Nyquist frequency).

The periodogram and why it is noisy

The naive estimator of $S(f)$ is the **periodogram**: take the discrete Fourier transform $\hat{z}(f) = \sum_n z_n e^{-2\pi i f n \Delta t}$ and form $|\hat{z}(f)|^2$ (suitably normalised). It is asymptotically unbiased but *not consistent*: its variance does not shrink as the record gets longer: a longer signal gives more frequency points, each still as noisy as before. Plotted directly it is a spiky, unreadable curve, from which one cannot reliably compare two channels.

Welch's method: segment, window, average

Welch's method turns the noisy periodogram into a usable estimate by *averaging*. The recipe is:

- (i) **Segment.** Split the record into several shorter, overlapping segments.
- (ii) **Window.** Multiply each segment by a smooth taper (a Hann window) that falls to zero at its ends. This suppresses the spurious high-frequency power (*spectral leakage*) that an abrupt segment boundary would otherwise inject.
- (iii) **Average.** Compute the periodogram of each windowed segment and average them.

Averaging K segments cuts the variance of the estimate by roughly a factor K , at the cost of coarser frequency resolution (set by the segment length rather than the whole record). The result is a smooth PSD estimate whose *level* can be compared meaningfully between two signals; it is the standard, robust spectral estimator. Each segment is typically detrended before transforming (e.g. by removing a linear fit), so that the estimate reflects the fluctuations about the trend, where the sensor noise lives, rather than the trend itself (implementation details: Sec. 2.6.1).

How these spectra read for the two altitude channels, and what the comparison implies for the choice of channel, is discussed where that choice is made (Sec. 2.7.1); this appendix covers only the method.

Implementation and Computational Details

Chapter 1

Overview

This part collects the implementation- and computation-level detail behind the choices made in the main text: the exact code module and script that produced each figure or table, the configuration values in force, sample sizes and computation times, and the status of each piece (implemented and tested, designed but not yet built, or planned). It is organised as a document within the document: its own chapters mirror the body chapter by chapter, so that a detail removed from the body sits in the corresponding chapter or section here. It is not meant to be read start to end – the body links to the relevant place at the point where a detail was trimmed out of the text, via a cross-reference such as “(implementation details: Sec. 3.5.2)”.

Contents of this part

1	Overview	44
2	The dataset	46
2.1	Data acquisition	46
2.1.1	Pipeline and tooling	46
2.1.2	Reproducibility	46
2.1.3	Integrity	47
2.2	The IGC tracklog format	47
2.3	Flight metadata and the catalog	47
2.4	Glider categories	47
2.5	Coverage and statistics	47
2.6	Trajectory pre-processing	47
2.6.1	Choice of altitude channel	47
2.6.2	Fix-level cleaning	49
2.6.3	Trimming of ground phases	54
2.6.4	Flight-level filtering	54
2.6.5	From geographic to local Cartesian coordinates	54
2.6.6	Uniform sampling within a flight	55
2.6.7	Smoothing and differentiation	56
2.7	Preliminary characterization	56
3	Next steps	57
3.1	Observables on the whole trajectory	57
3.2	Segmentation into flight phases	57
3.3	Phase-resolved observables	57
3.4	Reproducing the reference results	57

3.5	Transport analysis	57
3.6	Modelling and simulation	57
3.7	Tooling	57

Table 1.1 is a quick index of which script reproduces which auto-generated figure or table (all paths relative to `scripts/reporting/`); the chapters that follow go into the reasoning and numbers behind each one.

Figure/Table	Script	Notes
Stats macros; Tables 2.1, 2.2	<code>generate_stats.py</code>	Reads committed <code>data/*/seasons_index.csv</code> ; instant; part of the pre-commit hook.
Census macros (<code>\StatScan...</code>)	<code>generate_census_stats.py</code>	Reads the cached track scans on the SSD (never rescans); instant; best-effort in the pre-commit hook, so the committed macros refresh whenever the SSD is mounted.
Figure 2.1	<code>generate_altitude_noise_figure.py</code>	Needs the raw IGC archive (env. <code>SOARING_PARA_DATA_ROOT/SOARING_DELTA_DATA_ROOT</code>).
Figures 2.2, 2.3, 2.6, 2.7	<code>generate_preproc_figure.py</code>	One run produces all four (a shared full-census scan feeds three; fix-level cleaning uses a separate sampled pass). Needs the raw IGC archive.

Table 1.1: Script that regenerates each auto-generated thesis figure or table. Run as `uv run python scripts/reporting/<name>`; only `generate_stats.py` is wired into `scripts/build_docs.sh` (the two figure scripts need the external SSD and are run manually).

Chapter 2

The dataset

Mirrors Chapter 2 section by section, opening with the data acquisition (Sec. 2.1) and including the pre-processing and preliminary- characterization sections (Secs. 2.7, 2.8) – the cleaning and first-look steps are part of describing the dataset, not of the planned analysis in Chapter 3.

2.1 Data acquisition

Mirrors Sec. 2.1.

2.1.1 Pipeline and tooling

Package `soaring.acquisition.ffv1`, exposed as two CLI entry points, `soaring-para` and `soaring-delta`, sharing subcommands: `fetch-xml` (archive season XMLs), `download` (fetch IGC files), `build-catalog` (rebuild catalog/index), `status` (per-season coverage), `verify` (post-hoc integrity scan, Sec. 2.1.3).

Resumable downloads. Files already present on disk are skipped, so an interrupted job resumes freely without re-fetching.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an A header record followed by at least one B fix record); HTML error pages and truncated responses are rejected and retried.

Courteous networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests avoid hammering the server; a browser TLS fingerprint is impersonated to pass the site’s Cloudflare challenge.

2.1.2 Reproducibility

Destination directory (`data_root`) set per machine via an environment variable – `SOARING_PARA_DATA_ROOT` for paragliders, `SOARING_DELTA_DATA_ROOT` for hang gliders – or a config file shipping a placeholder (`configs/para_download.yaml/ configs/delta_download.yaml`); every command validates the setting at start-up.

2.1.3 Integrity

The `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors.

2.2 The IGC tracklog format

The parser (`soaring.analysis.igc.parse_igc`) checks, at decode time and independently of any flight-dynamics judgement: that the time is a valid 24-hour `hh:mm:ss`; that the arc-minutes field of latitude and longitude is < 60 (the `DDMM.mmm` encoding is undefined otherwise); that the hemisphere letter is one of the two valid characters; and that the decoded latitude/longitude fall within $[-90, 90]/[-180, 180]$ degrees. A record failing any of these is dropped as unparseable, on the same footing as a record with the wrong length or a non-numeric field. This format-validity check is implemented now and covered by unit tests; the fix-level dynamics-plausibility checks it is distinct from (Sec. 2.6.2) are designed and thresholded but not yet implemented.

2.3 Flight metadata and the catalog

Both `catalog.csv` and `seasons_index.csv` are regenerated from the archived season XMLs on demand, via the acquisition CLI's `build-catalog` subcommand (Sec. 2.1.1) – never hand-edited.

2.4 Glider categories

Label normalisation (not yet implemented) will merge `aile_class` variants and recover a blank or placeholder (“0”) class from the wing model (`aile`) where possible, at the fix-level/pre-processing stage (Sec. 2.6.2).

2.5 Coverage and statistics

To be populated: season-table provenance.

2.6 Trajectory pre-processing

Mirrors Sec. 2.7 and its subsections below. The Welch/PSD computational procedure used to compare altitude channels (Sec. 2.6.1) is the same one Sec. 2.7.7 will reuse, applied to E, N, U instead of altitude alone; 2.6.7 should cross-reference 2.6.1 for that shared procedure rather than restate it, once populated.

2.6.1 Choice of altitude channel

PSD sample (Figure 2.1a–c). A seeded random subsample of 3,000 flights per discipline – larger than a spectral *shape* alone would need, because the high-frequency barometric curve is visibly under-averaged (ragged) at a few hundred flights and smooths out only with a few thousand. Panel (a)/(b) use one representative flight (the one with the most fixes in the subsample); its measured offset (panel b) is annotated on the panel itself, not restated here.

Welch/PSD computational procedure (panel c). IGC loggers record one fix per second nominally, but timestamps are not guaranteed to be perfectly regular (a missed fix, a logger writing at 1.05 s instead of 1.00 s); Welch’s method (Appendix B) requires a uniformly sampled signal, so the following runs on every flight before the transform.

- (i) **Modal sampling period.** For each flight the median inter-fix interval is computed and rounded to the nearest integer second. The value that appears most often across all flights in the subsample (the *mode*) is taken as the common target Δt (in practice $\Delta t = 1$ s for both disciplines, giving $f_s = 1$ Hz and $f_{\text{Nyq}} = 0.5$ Hz).
- (ii) **Flight selection.** Only flights whose own median period lies within 0.25 s of the modal Δt are retained, so pooled spectra share one frequency axis; flights with fewer than $N_{\text{seg}} = 256$ fixes (~ 4 min at $\Delta t = 1$ s) are also excluded, since at least one full segment is needed. Of the 3,000 sampled flights per discipline, this leaves 1,641 paragliders but 727 hang gliders: paragliders are predominantly 1 s loggers (69 %, Sec. 2.6.6), so the population mode is a good match for most of them, whereas hang gliders split across several native rates (1, 2, 5, ≥ 8 s, Figure 2.7) and only their 1 s-logger minority survives this cut.
- (iii) **Uniform resampling.** The altitude series $z(t)$ is linearly interpolated onto a grid $t_0, t_0 + \Delta t, t_0 + 2\Delta t, \dots$ spanning the flight, without introducing frequency content beyond f_{Nyq} .
- (iv) **Welch on the resampled series.** `scipy.signal.welch` is called with $f_s = 1/\Delta t$, $N_{\text{seg}} = 256$, 50% overlap, a Hann window, and linear detrending per segment.
- (v) **Robust ensemble estimator.** The per-flight PSDs are kept individually (per discipline and channel) and reduced across flights by the per-frequency *median*, with a shaded 10 to 90 percentile band; panel (c) then pools the two disciplines, whose medians coincide. The median, not the arithmetic mean, because these per-flight spectra are heavy-tailed: a mean is dominated by a few flights and swings by one to two orders of magnitude between samples – for the hang-glider barometric floor, from 0.8 to 93 m² Hz^{−1} between an $n = 114$ and an $n = 727$ draw – whereas the median is stable. Because Δt is the same for every retained flight by construction, the figure’s Nyquist dotted line is a single fixed value.

Reading the band (panel c).

Why the medians coincide. For the typical flight both channels are dominated at high frequency by the metre resolution to which the IGC format rounds every altitude: the white-noise floor of a 1 m quantiser, $\frac{1}{12}/f_{\text{Nyq}} \approx 0.17$ m² Hz^{−1}, sits just below where the measured medians level off (≈ 0.3 m² Hz^{−1}).

Where the channels part. In the tail: at the 90th percentile the high-frequency GNSS floor is 4× the barometric one for paragliders and 108× for hang gliders (the worse GNSS multipath of that fleet, and the reason its band is wider), and 13 % to 20 % of flights carry a GNSS floor above three times their own barometric one.

Consequence. This heterogeneity, rather than a uniform gap, motivates the choice of the barometric channel (Sec. 2.7.1).

Fallback-rate census (panel d). A full census, not a sample: the same full parse of every track is needed anyway for the flight-level filtering diagnostics of Sec. 2.6.4 (Figure 2.3), so the barometric-presence fraction is read off that one scan rather than measured separately, at no extra cost. (A sized random sample – the standard proportion formula $n = z^2 p(1 - p)/e^2$ – would be the right tool if a full census were not already being paid for elsewhere; it is not needed here.)

Presence is whole-file (panel d). The barometric channel is all-or-nothing at the file level, checked on the same census: when it is absent it reads exactly zero for 99.7% (paragliders) / 100% (hang gliders) of those flights, and when present it covers every fix for 96% / 98% of them (the rest miss only a handful of individual fixes, dropped one by one, Sec. 2.7.2). A logger either has a pressure sensor for the whole track or none at all, so the `alt_source` decision is per flight, never per fix.

2.6.2 Fix-level cleaning

This section gives the working definitions the narrative (Sec. 2.7.2) states in words: the three detectors, their parameters, the order they run in, and how the whole step is validated. The three families of Table 2.3 map onto three detectors of increasing context: an absolute bound (no context), a robust local test (a short window of context), and a structural rule (a whole run).

Status. The format-validity rejection is implemented in `soaring.analysis.igc.parse_igc` and covered by unit tests. The dynamics and structural cleaning below is designed, and its absolute bounds are thresholded on the data (Table 2.3, Figure 2.2); the routine that walks a trajectory applying all three detectors, and the robust-test and frozen-lock parameters it needs, are the next step. The parameter names introduced here are the config keys that step will add.

Robust local-outlier test (off-the-trend defects). For a fix P_k at time t_k :

- (1) take the fixes lying within $\pm w$ seconds of t_k (a *time* window, so it adapts to the native cadence and to irregular sampling), excluding P_k itself;
- (2) fit a robust local estimate of the position there – the component-wise median of the windowed fixes, or a degree-one Theil–Sen fit – and evaluate it at t_k ;
- (3) the residual r_k is the great-circle distance from P_k to that estimate; scale it by the robust local spread $\sigma_k = 1.4826$ MAD of the windowed residuals (the factor makes σ_k a Gaussian-consistent standard deviation [16]);
- (4) flag P_k when $r_k > k \sigma_k$ and $r_k > \varepsilon_{\min}$.

This is a Hampel identifier [6]: a median centre, a MAD scale, a threshold in robust standard deviations; working defaults in Table 2.1. The window must hold at least a handful of fixes (order five); a sparser neighbourhood falls back to the absolute bounds alone.

Three points earn the test its place over the bare speed bound:

- *The floor $\varepsilon_{\min}$ is not cosmetic.* On a very smooth stretch $\sigma_k \rightarrow 0$, so without it any sub-metre wiggle would exceed $k\sigma_k$ and the test would chase GPS noise where there is no outlier.
- *It is local, and locality buys attribution.* A speed belongs to the segment *between* two fixes and cannot say which endpoint is at fault, whereas the residual belongs to one fix: when the absolute bound fires, the identifier attributes the defect to a single fix, and a

burst is removed as one block – a short burst of consecutive spikes inflates the windowed MAD only mildly, so it is still flagged, and is removed as the block whose bracketing neighbours rejoin within scale.

- *It does not clip real manoeuvres.* A genuine sharp turn is drawn by consecutive, mutually consistent fixes, each well predicted by its neighbours and so of small residual; only an isolated point off the local trend scores large. The one failure mode – a turn so tight it is under-sampled at the native cadence – is rare at these rates, and the injected-defect test below measures it directly.

The flag itself is context-sensitive – a 30 ms^{-1} step out of a 12 ms^{-1} glide is flagged while the same step passes unremarked where it is ordinary – but a flag alone does not delete (next paragraph): a flagged-but-physically-possible fix is *kept*, its flag recorded as a per-flight anomaly count alongside the integrity-gate counters. The identical test runs on the horizontal position and, independently, on the adopted altitude channel, so a vertical spike invalidates the altitude alone and the horizontal position stays (the channel decoupling of Sec. 2.7.2).

From flag to action. The identifier only *flags*; it never repairs. A flagged horizontal fix is *deleted* as a node – not overwritten with the local median – so that all reconstruction is done by the single resampling step (Sec. 2.6.6), under one policy and one accounting, rather than by a second, in-place imputation that would then feed the resampler as if it were data. This is the *Hampel identifier* [6], the detection half of the eponymous filter, not the median-substituting filter itself: the local median/Theil–Sen enters only as the reference for the residual r_k , never as an output value.

Position. Deletion requires two conditions: the fix must be flagged *and* its implied in-and-out speed must exceed the absolute v_{xy} bound. The gate narrows the identifier’s deletions to what the bound already deems impossible; the one benign false positive, a genuine under-sampled tight turn (flagged, but flying at ordinary speed), is kept, and its bounded residual effect is absorbed by the smoother.

Discontinuity. When no bounded block restores continuity – a position discontinuity, such as a re-acquisition offset, where every removal still leaves an impossible step – the scan stops deleting and marks a split at the step, handled like a long gap (Sec. 2.6.6): both sides are genuine, only the transition is unknown.

Altitude. A flagged altitude is marked missing on the flag alone: a dropped altitude is a deferral, restored by the audited interpolation at resampling, so the stakes are far lower than for a position node. On a GNSS-fallback flight the absolute $|v_z|$ bound (calibrated on the barometric channel) does not apply, and this test is the only vertical cleaner.

This is the *default to keep* rule of Sec. 2.7.2: removal requires positive, corroborated evidence.

Frozen-lock detection (on-the-trend). A run is a lost-lock episode only under the rule of Eq. (2.3) – spatially collapsed, *and* witnessed (barometrically flat or byte-identical; the declared flags only where no barometer exists), *and* long enough – and the pass defaults to keeping the data otherwise.

- *Extent.* The run’s bounding diameter – the maximum great-circle distance between *any* two of its fixes, not the fix-to-fix step – stays below ε (a few times the GPS noise, $\sim 10\text{ m}$ to 15 m); a circling climb spans its disc, $2R \approx 40\text{ m}$ to $100\text{ m} \gg \varepsilon$, and is cleared on extent alone, whereas a step-based test would fire on every 1 Hz climb (step $v_{xy}\Delta t \approx 10\text{ m} \sim \varepsilon$).

- *Independent witness.* Over the run the barometric altitude must also be flat ($|\Delta z_{\text{baro}}|$ below a few metres), since a real thermalling climb gains tens of metres in that time on a sensor independent of the GNSS, so a horizontally-still run whose baro altitude is still climbing is a tight climb and is kept. Δz is taken robustly (medians of the run’s ends), so an altitude spike inside the run – not yet cleaned, since the altitude pass comes last – cannot fake a climb.
- *Corroboration.* Of the declared signatures, only the *byte-identical* repeat may override the witness – a run of byte-identical coordinates under a climbing barometric altitude is a freeze that happened *during* a climb (no real circling wing rewrites the same coordinates), and is cut, the positions being invented even where the height gain is real. The V flag and the zeroed GNSS altitude do *not* override a live barometer: they certify a degraded GNSS *altitude*, not a frozen horizontal fix (a V fix may still carry a usable position, Sec. 2.2), so on a barometric flight they are recorded as corroborating diagnostics only, and become decisive only on a GNSS-fallback flight, where Eq. (2.3) substitutes them for the missing barometric witness.
- *Duration.* The run must span at least τ_{freeze} , set to a few thermalling periods ($\gtrsim 60$ s to 90 s), so that a long, slow, tight climb is never mistaken for a freeze.

Only then is the run marked as a data gap; the split it induces is realized at the uniform-sampling stage (Sec. 2.6.6), on the same footing as a native long gap. On a GNSS-fallback flight the barometric witness is unavailable, so there the extent and flag tests must both hold outright and, in any doubt, the run is kept – the safe direction, since keeping a rare true freeze costs far less than splitting a real climb.

Duplicate and backward timestamps (on-the-trend). The parser already rebuilds a monotone clock: it unwraps the UTC-midnight roll-over (a drop of $\sim 86,400$ s advances the day count) and clamps residual backward jitter with a running maximum (Sec. 2.2). The redesign changes one behaviour: a backward step that is *not* a roll-over is no longer silently clamped but removed – the fix with the corrupt time is deleted as a node, just as a fix with a corrupt position is. A corrupted timestamp is recorded as a defect rather than silently repaired. Which fix carries the corrupt time is decided by minimal removal: delete the smallest set whose removal leaves a strictly increasing clock (the complement of the longest increasing subsequence), so a single fix with a forward-jumped clock is itself removed, rather than condemning every later fix against a running maximum. This requires moving the clamp out of the parser: `parse_igc` keeps the roll-over unwrap (a format concern, needing the raw time of day) but stops flattening backward jitter with the running maximum – flattened, the evidence the cleaning must act on no longer exists downstream.

Fixes sharing one UTC second are collapsed to a single representative at that second’s centroid position. This replaces the provisional “keep the first” of the earlier design: keeping the first privileges an arbitrary member and, for a genuine ≥ 2 Hz logger, would silently halve the cadence, whereas the centroid is order-independent and the sub-second resolution it discards is discarded anyway once the flight is resampled to a common Δt (Sec. 2.6.6). One dilution case is accepted: if one member of a duplicate pair is itself a position spike, the centroid halves its amplitude before the outlier pass (which runs after the time defects) sees it; a large spike is still flagged at half size, and one small enough to slip under $k\sigma_k$ after halving is within what the smoother absorbs.

Absolute bounds (out-of-range, the backstop). The six numbers of Table 2.3 are the `FixLevelThresholds` dataclass, loaded by `soaring.analysis.preprocessing.load_preproc_config` from `configs/preprocessing.yaml` (key `fix_level`); nothing is hard-coded

in Python. They are the context-free floor of the design, and are placed on the real per-fix distributions (Figure 2.2) rather than guessed. The horizontal-speed bound is set where the bin-to-bin ratio of a finely binned (2.5 m s^{-1}) histogram settles at ≈ 1 , i.e. where the decay has actually stopped, not where it first visibly slows ($\approx 40 \text{ m s}^{-1}$ for hang gliders, versus the adopted 55 m s^{-1}), since a shallow but real decreasing stretch can still be genuine rare dynamics.

Diagnostic sample (Figure 2.2). Computed by `fix_level_distributions` and rendered by `make_fixlevel_diagnostics_figure` (Table 1.1). Panels (a)–(c) pool a seeded random sample of 15,000 paraglider flights (126,723,499 consecutive-fix pairs, $\sim 73 \text{ s}$ to compute) and the complete 6,716-flight hang-glider population (a full census at this size, 37,915,334 pairs, $\sim 25 \text{ s}$) – under two minutes total. The large sample is deliberate: resolving the sparse artefact tail well enough to tell it from real dynamics needs far more points than a headline statistic would. The x -range in (a)/(b) extends to each discipline’s own 99.99th percentile rather than cropping just past the bound; panel (c) keeps a narrower range, its cut sitting far out in an otherwise tight, unimodal distribution. Panel (b) bins one integer m s^{-1} per bin because barometric altitude is logged to the metre: differencing at a 1 s cadence returns integer rates, and a finer grid would alias them into a spurious comb. (Not every flight samples at 1 s, Figure 2.7, and differencing over a longer native step returns finer values (multiples of 0.2 m s^{-1} at 5 s) that partly fill the gaps once cadences are pooled: among 1 s-native fixes about 99% land within 0.05 m s^{-1} of an integer, against only 34–42% at every other cadence.)

Detector order and complexity. The detectors run in a fixed order:

- (1) format validity, at decode time (Sec. 2.2);
- (2) time defects: backward steps, then duplicates;
- (3) position outliers (Hampel flag + impossibility gate);
- (4) frozen-lock runs;
- (5) altitude: absolute bounds and altitude outliers.

The order matters because a speed or a residual is only meaningful once the time base is monotone and free of duplicates. The position-outlier and frozen-lock passes are one left-to-right scan that carries the last accepted fix as an anchor: it never re-walks the whole track, and it absorbs a run by extending the anchor across the flagged block – tracking the block’s running bounding box to test its extent against ε – so the cost is $O(N)$ in the number of fixes.

Flight-level integrity gate. The pass records how much it changed each flight (or segment) – `n_fix_raw`, `n_fix_clean` and `frac_interpolated` in `flights_meta` – counted over the airborne (trimmed) window only: junk in a ground phase that trimming removes anyway must not condemn the flight. A flight whose deleted-plus-reconstructed fraction exceeds a small f (order 0.05 to 0.10) is dropped whole: beyond that point the trajectory is more interpolation than measurement. This is the cleaning counterpart of the sampling-regularity cut (Sec. 2.6.6).

Configuration. The absolute bounds are already the `FixLevelThresholds` group above. The remaining parameters are collected in Table 2.1: they join that group in `configs/preprocessing.yaml` when the routine is built and, as with every threshold in this pipeline, they live in the config, never in Python.

Validating the cleaning. The removed-fraction count of Figure 2.2 bounds only the false positives (over-cleaning). Five checks, part of the same planned step, bound the false negatives and the bias:

Parameter	Working default	Role, and the scale that sets it
<i>Robust local-outlier test (Hampel)</i>		
w	20 s	half-width of the local time window; a few native intervals
k	5	flag threshold, in robust standard deviations σ_k
$\varepsilon_{\min}$	≈ 15 m	absolute residual floor; a few times the horizontal GPS noise, so a smooth stretch ($\sigma_k \rightarrow 0$) is not chased
<i>Frozen-lock rule (Eq. 2.3)</i>		
ε	15 m	run-extent tolerance; a few times the GPS noise, $\ll 2R \approx 40$ m to 100 m of a circling climb
δ_z	≈ 5 m	barometric-flatness witness; a few times the sensor noise and quantisation, $\ll$ the $\gtrsim 60$ m a real climb gains over τ_{freeze}
τ_{freeze}	60 s	minimum span of a cut run; two thermalling periods, so a slow, tight climb is never mistaken for a freeze
<i>Flight-level integrity gate</i>		
f	0.10	maximum deleted-plus-reconstructed fraction; past it the flight is being rebuilt, not cleaned, and is dropped whole

Table 2.1: Fix-level cleaning: parameters of the robust and structural detectors, with their working defaults – to be finalized by the injected-defect calibration, check (1) of the validation list below – and the physical scale that sets each. They live as config keys under `fix_level` in `configs/preprocessing.yaml`; the absolute bounds of Table 2.3 already live there.

- (1) *Injected defects.* On tracks the cleaning leaves untouched, plant synthetic defects of known type and size – position spikes of amplitude a , frozen runs of length τ , backward-time glitches – and measure the recovered fraction against size. This gives the detection floor (the smallest spike caught) and calibrates k , w , ε and τ_{freeze} to a target rate: the only direct measurement of what the cleaning misses.
- (2) *Downstream invariance.* Recompute the transport estimators – the displacement-scaling exponent and the waiting-time-tail exponent – over a grid of cleaning strengths, and take the operating point on the plateau where they stop moving; movement under stricter cleaning is the signature of a cut biting into signal.
- (3) *Detector nesting.* The fixes flagged by the absolute bound should nest inside those flagged by the local test (the impossible is a subset of the locally anomalous); a large disagreement points to a defect class neither was designed for – a cheap guard.
- (4) *Per-phase removal rates.* The error this pipeline most wants to avoid is losing a slow-phase fix. Once the segmentation is in place (Ch. 3), the removal and split rates are measured *per phase* and required to be flat across transition, search and climb rather than enriched in the two slow phases; and synthetic tight climbs – circles at flying speed, over a range of radii and wind drift – planted in otherwise clean tracks confirm directly that the frozen-lock detector never splits a real climb. This is the false-*positive* complement of check (1), and its natural readout is the tail of $\psi(\tau_w)$: a cut biting into climb or search

would thin that tail before it moved anything else.

- (5) *Isolation of the vertical cuts.* The fixes cut by $|v_z| > 13 \text{ ms}^{-1}$ must be *isolated*, not consecutive: a sustained spiral dive holds 15 ms^{-1} to 20 ms^{-1} of real sink, so a run-shaped excess is the signature of a genuine manoeuvre – calling for a raised bound or a coherent-run exemption rather than censoring – whereas an isolated super-threshold difference, bracketed by ordinary rates, is a sensor error and safe to cut.

Reproduce. Table 1.1, row `generate_preproc_figure.py` (one run also regenerates Figures 2.3, 2.6 and 2.7 from a shared full-census scan; the fix-level panels use a separate sampled pass).

2.6.3 Trimming of ground phases

The take-off speed (5 ms^{-1}) and sustained duration (30 s) are the `TrimmingThresholds` dataclass, loaded from `configs/preprocessing.yaml` (key `trimming`); as with every threshold in this pipeline, nothing is hard-coded in Python.

The interior-ground guard of Sec. 2.7.3 keeps `interior_ground_s` (T_{ground} , 600 s) and `ground_flatness_m` (5 m) in the same group. A detected stint is excised and handed to the split machinery of Sec. 2.6.6, so it needs no removal semantics of its own. Below T_{ground} the same predicate is stored as a list of suspect intervals per flight; the segmentation later maps them onto the phases they touch, and the $\psi(\tau)$ sensitivity re-fits consume the flag.

2.6.4 Flight-level filtering

The duration window ($40 \text{ min} \leq T \leq 16 \text{ h}$), the path floor ($L \geq 10 \text{ km}$) and the altitude-activity floor (75 m on the adopted channel) are the `FlightLevelThresholds` dataclass, from `configs/preprocessing.yaml` (key `flight_level`); the census measurements behind the 30-to-10 km change and the upper duration bound are in the narrative (Sec. 2.7.4).

Diagnostic sample (Figure 2.3).

Scope. The full census, every parsed track, no sub-sampling: 186,025 paraglider and 6,716 hang-glider tracks, from the `load_or_scan_tracks` scan; the rendering script is in Table 1.1.

Provenance. The totals, like every census number in this document, are generated macros quoting the cached scan (`generate_census_stats.py`); the live archive grows by a trickle, so they can trail the download counts of the season tables until the cache is refreshed.

Headline medians. Duration and path length: 2.6 h/86 km for paragliders, 3.1 h/158 km for hang gliders; the latter fly faster and farther.

One pass. The scan is cached at `<data_root>/derived/track_scan.parquet` and also feeds Figures 2.6 and 2.7.

Reproduce. Table 1.1, row `generate_preproc_figure.py`.

2.6.5 From geographic to local Cartesian coordinates

No implementation content: this step is pure coordinate geometry with no configurable parameters, and both of its figures (Figure 2.4, Figure 2.5) are hand-drawn TikZ diagrams, not data plots.

2.6.6 Uniform sampling within a flight

The sampling thresholds (key `sampling`, the `SamplingThresholds` dataclass) act as follows.

Split bound. $g_{\max} = \min(\text{max_gap_factor} \times \Delta t, \max(\text{max_gap_seconds}, 2 \Delta t))$, pinned by the constraint chain of Sec. 2.7.6. Its audit is a per-flight count – the census scan records `max_gap_ratio` and `dt_s`, so the absolute gap is their product – and joins Figure 2.6 when regenerated.

Missing fraction. Acts per *segment*: a segment more than a tenth uncovered is dropped alone, the flight only if no segment survives. The per-flight value in the census scan is the pre-split diagnostic, not the operational gate.

Segment gate. `min_segment_duration_s` (90 s). It only needs to support the smoothing window and the within-segment statistics: a few smoothing timescales, never less than $w \Delta t$.

Measured vs interpolated. A grid point counts as measured when a fix lies within half a native step of it, interpolated otherwise; the same tolerance decides whether a flight is uniform as recorded (no interpolated points) or was resampled (`was_resampled`).

The bound is on the *time base* and is the same for every channel: an order-of-magnitude smaller v_z buys no wider gap for the altitude, since the interpolation error scales with curvature, not speed, and a wide altitude gap would erase the climb–glide transitions that v_z is there to resolve (Sec. 2.7.6).

Segment handling.

Origin and clock. A segment keeps its parent flight’s origin and clock. The ENU conversion precedes this step, so every fix already carries parent-referred coordinates and time; a split only assigns a `segment_id`.

Gates. The flight-level cuts are not re-applied to a segment (Sec. 2.7.6); only the minimal-duration gate above applies, and a dropped segment does not drop its siblings.

Tables. One row per fix in `fixes`, keyed (`flight_id`, `segment_id`); per-segment records (start/end, fix count, `frac_interpolated`, censoring flags) in a `segments` table beside `flights_meta`. No separate files; the ensemble MSD at lag t reads $\mathbf{r}(t)$ wherever some segment covers t .

Censoring. Any phase duration truncated at a segment boundary is flagged censored, for the $\psi(\tau)$ fits to exclude.

Per-channel fill. Across a short gap the horizontal position is interpolated shape-preservingly (monotone piecewise-cubic, e.g. `scipy.interpolate.PchipInterpolator`) and the barometric altitude linearly or by cubic spline (its within-phase smoothness makes the choice immaterial). The result meets the completeness invariant of Sec. 2.7.2: within each retained segment every grid time carries a defined (x, y, z) – deleted position nodes and deferred (missing) altitudes both restored here – so the segmentation is never handed a hole; long gaps are split, not filled. Each filled grid point is flagged in an `interpolated` boolean column of `fixes`, and `frac_interpolated` records the per-segment fraction for the integrity gate.

Figures 2.6 and 2.7. Both rendered by `make_gap_diagnostics_figure/make_sampling_figure` from the same full-census scan as Figure 2.3 (Table 1.1, row `generate_preproc_figure.py`) – no separate parse needed.

Gap-ratio discreteness (Figure 2.6). Checked directly: only 86%/63% (para/hang) of largest-gap-ratio values are exact integers, the rest a genuinely continuous scatter; the integer peaks visible for hang gliders sit around $k = 3, 6, 10, 15$.

Native-rate discreteness (Figure 2.7). Checked directly: Δt lands on an exact whole second for 99.9% of paragliders and 100% of hang gliders. Per-rate breakdown: paragliders mostly (69%) at 1 s; hang gliders spread across 1 s (30%), 5 s (27%), 10 s (15%), 2 s (13%), and a long thin tail beyond 11 s.

2.6.7 Smoothing and differentiation

Not yet implemented. Three decisions are pinned for the implementation: the filter runs per segment (never across a boundary) with `scipy.signal.savgol_filter` in `mode='interp'`, so the terminal half-windows are fitted on the edge window rather than padded; edge samples inherit the segment-boundary caution of Sec. 2.6.6; and the window is floored at $w = 5$, the smallest a cubic fit accepts, which for the few flights with $\Delta t \gtrsim \tau_c$ smooths slightly more than the noise scale (those flights are stratified out of the Δt -sensitive observables anyway). The polynomial order ($p = 3$) and the three smoothing timescales – `tau_c_horizontal_s` for (E, N) , `tau_c_vertical_baro_s` and `tau_c_vertical_gnss_s` for U – are the `SavgolParams` dataclass in `configs/preprocessing.yaml` (key `savgol`).

The measured timescales. The knees were measured with `estimate_savgol_timescales.py`, which applies the Welch procedure of Sec. 2.6.1 to E, N (local equirectangular metres) and to the altitude channel each flight actually uses, on a seeded sample (908 horizontal flight-components, 719 barometric and 189 GNSS-fallback flights at 1 Hz). The knee rule: the floor is the median PSD over 0.35 Hz to 0.5 Hz; f_c is the lowest frequency whose median PSD falls below three times it. All three knees land at $f_c \approx 0.2$ Hz, so $\tau_c = 5$ s for every channel: each channel’s typical floor is the metre and arc-minute quantization of the IGC format (measured ≈ 0.60 and $\approx 0.26 \text{ m}^2 \text{ Hz}^{-1}$ against the theoretical $q^2/12/f_{\text{Nyq}}$ of 0.57 and 0.17), not receiver noise. The expected asymmetry between the barometric and GNSS verticals shows in the spectrum’s *level*, not in its knee, and even the 90th-percentile spectra move the knee by less than a second. The two vertical keys stay, conditioned on `alt_source`, for the noisy-minority refinement; the validation of Sec. 2.7.7 (flat residual spectrum, stability under one window step) is what confirms the values. The filter runs on the resampled grid, so the small interpolated fraction (`frac_interpolated`) is already in place when it reads its window.

2.7 Preliminary characterization

Not yet implemented; see Sec. 2.8.

Chapter 3

Next steps

The sections below mirror Chapter 3, which – since pre-processing and preliminary characterization now live in Chapter 2’s mirror (Sec. 2) – starts from “Observables on the whole trajectory.” None of this analysis code exists yet, so each entry is a placeholder rather than a populated record; each will gain its own implementation content (module names, sample sizes, reproduction commands) as that code is written.

3.1 Observables on the whole trajectory

Not yet implemented; see Sec. 3.1.

3.2 Segmentation into flight phases

Not yet implemented; see Sec. 3.2.

3.3 Phase-resolved observables

Not yet implemented; see Sec. 3.3.

3.4 Reproducing the reference results

Not yet implemented; see Sec. 3.4.

3.5 Transport analysis

Not yet implemented; see Sec. 3.5.

3.6 Modelling and simulation

Not yet implemented; see Sec. 3.6.

3.7 Tooling

Largely underway already: see Sec. 3.7 and the reproducibility map, Table 1.1.